

The temporary-context theory of conditionals

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Abstract

This paper proposes a minimal, but powerful, theory of indicative conditionals built around three components: a theory of local contexts; the idea that *if*-clauses function to create a temporary context (and do nothing else); and the claim that worlds outside the context of evaluation are treated, not as true or false, but as irrelevant to semantic evaluation. Conditional restriction of modals emerges as an automatic byproduct of the creation of temporary contexts, and the theory explains parallels between presupposition projection and domain restriction in both conditionals and disjunction. It also accounts for Stalnaker’s Thesis, the equation between probabilities of conditionals and conditional probabilities. By deriving a trivalent semantics from more basic components, the temporary-context theory answers an explanatory challenge for trivalent theories of conditionals and points to a novel generalization: possibilities that lie outside the context of evaluation are systematically invisible to natural language interpretation.

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1 Background and overview

This paper explores the claim that everything about indicative conditionals can be derived from the following thesis, in interaction with independently motivated mechanisms.

If-clauses are instructions to create a **temporary context**.

That is, *if*-clauses manipulate the context that they are handed, and do nothing else. For indicative conditionals, this means that *if A* temporarily restricts the context of evaluation—viewed as a set of worlds (Stalnaker, 1978)—to the worlds where *A* is true. This manipulation is then passed along to whatever happens next, whether this involves assertion, embedding, or probability evaluation, and it affects presupposition satisfaction in predictable ways.

The intuition that conditionals involve temporary context manipulation has a long pedigree. Most famously, Ramsey (1929) writes: “If two people are arguing ‘If *p* will *q*?’ and are both in doubt as to *p*, they are adding *p* hypothetically to their stock of knowledge and arguing on that basis about *q*”. In a similar vein, Mackie (1973) argues that “[t]he basic concept required for the interpretation of ‘if’-sentences is that of supposing . . . To assert ‘If *p*, *q*’ is to assert *q* within the scope of the supposition that *p*”. Stalnaker (1968) concurs: to interpret a conditional, you should “add the antecedent (hypothetically) to your stock of knowledge and then consider whether the consequent is true. Your belief about the conditional should be the same as your hypothetical belief, under this condition”.

While the idea that an assertion of *if A, C* just is the assertion of *C* within a temporary, restricted context is widely thought to be compelling, few formal theories of conditionals have implemented it directly. Most theories of conditional assertion (e.g., Ducrot 1972; Mackie 1973) have been oriented toward a special-purpose speech act of “conditional assertion” that is widely thought to be mysterious and difficult to implement formally. Despite the promise of the quote given above, the semantic theory

proposed by Stalnaker (1968) does not actually involve context manipulation, but rather shifting of the world of evaluation. The influential restrictor theory of Kratzer (1981, 1991a, 1991b) comes close, positing that *if*-clauses manipulate a contextual parameter. However, the restriction involves a special-purpose parameter that is separate from the context, and Kratzer maintains the idea that indicative conditionals introduce an additional layer of modal or quantificational interpretation. This quantificational aspect is also shared by a number of recent theories, reviewed below, that resemble the temporary context theory in other respects (Carter, 2021; Klinedinst & Rothschild, 2012; Yalcin, 2007). As we will discuss below, this choice brings with it a number of problematic predictions about the truth-conditions and probabilities of conditionals.

By contrast, in the theory proposed here the only thing that an *if*-clause does is to manipulate a parameter representing the context set in the sense of Stalnaker (1978). Crucially, this parameter does not merely set the domains of modal operators, as in the restrictor theory: it also does a variety of further semantic and pragmatic jobs. As a result, the temporary-context theory unifies the domain-restricting effects of conditionals and other operators (notably disjunction) with their effects on the projection and filtering of presuppositions. The new version of the temporary-context theory proposed here enjoys a number of further theoretical and empirical advantages over competing theories. It avoids the need to posit otherwise unmotivated silent operators in bare conditionals, and it correctly predicts two phenomena that are challenging for many accounts: the fact that the assertive force of a bare indicative conditional is the same as that of a material conditional; and Stalnaker’s Thesis, the equation between probabilities of conditionals and conditional probabilities. Indeed, we will show that the theory correctly predicts a generalization of Stalnaker’s thesis to certain embedded contexts.

The main technical innovation of this paper is to combine a theory of local contexts with a trivalent valuation function which treats worlds outside the local context as *irrelevant* to the interpretation of a sentence. The theory of local contexts adopted here is derived from the literature on presupposition projection (e.g., Heim 1983; Karttunen 1974; Schlenker 2009). Our theory of conditionals takes inspiration from Kaufmann’s (2000) dynamic theory, though we simplify considerably by employing a static implementation based on Klinedinst and Rothschild 2012; Yalcin 2007.

Specifically, we show that the temporary-context approach is able to derive from first principles a trivalent, truth-functional theory of conditionals first proposed by de Finetti (1936) and recently revived and refined by Égré, Rossi, and Sprenger (2021); Huitink (2008) and others. The key idea of trivalent theories is that indicative conditionals with a false antecedent are neither true nor false: they have a third value $\#$ which can be interpreted as “irrelevant”. The temporary-context theory thus inherits a number of advantages of the trivalent semantics for conditionals, reviewed below, along with those of dynamic theories following Kaufmann 2000 and of the restrictor theory. The result is a unified theory of discourse-level features of conditionals, their compositional interactions with modals and other operators, and their relationship to presupposition, all emerging as by-products of local context manipulation.

At the same time, the temporary-context theory responds to a key explanatory problem for the trivalent theory of conditionals. While work in this area has demonstrated a number of theoretical and empirical advantages of the trivalent semantics, it has focused mainly on the technical issue of how the third truth-value can be put to productive use. Previous work has not addressed the question of *why* conditionals should be special, among the many construction types in natural languages, in invoking a third truth-value. This issue is crucial in situating the trivalent semantics within a complete theory of meaning for natural language. The temporary-context theory provides a straightforward answer: conditionals are not special. Instead, the way that conditionals employ the $\#$ value derives from the interaction of local context manipulation with a trivalent valuation function that applies to all expression types. This approach correctly predicts that other expressions that manipulate local contexts—e.g., disjunction and *supposing*-clauses—have predictably similar effects on presupposition, modal domains, and the like.

The temporary-context theory thus explains why conditionals are trivalent in terms of a very general property of natural language interpretation: possibilities that are excluded by the context of interpretation are always treated, not as true or false, but as irrelevant to the semantic computation. As we will see below, this feature explains why false-antecedent worlds are ignored in calculating assertive update, in computing the domains of modals and other operators, in calculating probabilities, and in the semantic evaluation of connectives and modals. Because they lie outside the (temporary) context of evaluation, these worlds are invisible to the semantics and pragmatics of language understanding.

2 Theoretical background

The temporary-context theory draws inspiration from a number of well-known formal theories of conditional semantics. This section briefly reviews the relevant aspects of these theories and their empirical and theoretical advantages, as well as lingering drawbacks. In the following sections I show that the temporary-context theory, by combining features of these various accounts, allows us to maintain their key advantages while avoiding the problems noted, and does so with minimal machinery specific to conditionals.

2.1 Strict conditional theories

Work on conditionals in philosophical logic has long been dominated by variants of the *strict conditional* theory introduced by C.I. Lewis (1912). Lewis argued that the truth-functional analysis of the conditional presented in *Principia Mathematica* (Whitehead & Russell, 1910) did not capture the intuitive notion of “implication”. Lewis’ objections—now well-known among the numerous “paradoxes of material implication”—largely revolve around the fact that a material conditional $p \supset q$ is true whenever p happens to be false, even if p and q are informationally unrelated. This feature does not seem to hold of *if*, as many authors have noted. In addition, the material conditional theory validates a number of argument forms that are clearly specious. For example, $\neg(A \supset C)$ is equivalent to $A \wedge \neg C$. The following argument is thus predicted to be valid, contrary to fact (Edgington, 1995).

- (1) a. It’s not the case that, if God exists, we will be punished for our sins.
b. Therefore, God exists and we will not be punished for our sins.

C.I. Lewis (1912 *et seq.*) argued that implication should be viewed instead as an intensional relation: p implies q iff it is “it is *impossible* under any circumstances that p should be true and q false” (C.I. Lewis and Langford 1932: 261). In modern intensional semantics and modal logic, this idea has been developed into a semantics for *if* in which it functions as a universal quantifier over a set of worlds $R(w)$ that are accessible from the evaluation world w .

- (2) $\llbracket \text{if } A, C \rrbracket^{R,w} = 1$ iff, for all $w' \in R(w)$, $\llbracket A \supset C \rrbracket^{R,w'} = 1$.

Here, *if* is treated as a universal modal requiring that, in every accessible world in which the antecedent is true, the consequent is as well.

In the strict conditional theory, the modal introduced by *if* quantifies over a domain that is fixed by the contextual accessibility relation and the evaluation world. An important development of the theory was the introduction of *variably strict* semantics, where the domain of quantification also depends on the antecedent A . The most influential variably strict theories are due to Stalnaker (1968) and D. Lewis (1973) (proposed for subjunctives only). Variants of these theories continue to be developed in the modern literature on conditionals in logic, philosophy, and linguistics.

For our purposes, the crucial features of strict-conditional theories are (i) that their primary motivation historically involved empirical failures of a bivalent, truth-functional semantics, and (ii) that they incorporate a universal quantifier in the lexical semantics of *if*. The first property will be important when considering motivations for a recently popular non-modal semantics based on trivalent valuations. The second means that the strict conditional theory encounters a number of empirical problems that have been much-discussed in subsequent literature.

Desideratum 1. Commutation with negation.

It is widely thought that the indicative conditional in natural language commutes with negation: that is, *If A, not C* is equivalent to the negation of *If A, C*. In support of this contention, note, for instance, that Bert’s contribution in (3b) is incoherent.

- (3) a. Al: If it rains, the picnic will be canceled.
b. Bert: No: it’s not true that, if it rains, the picnic will be canceled. But it’s also not true that, if it rains, the picnic will not be canceled.

In a strict conditional theory, (3b) should convey a perfectly coherent thought: that the relevant information fails to exclude either rain and a picnic, or rain and no picnic.

Commutation is closely related to, and implied by, the widely endorsed principle of Conditional Excluded Middle: at any world w , either *If A, C* or *If A, not C* is true. See [Cariani and Goldstein 2020](#); [Stalnaker 1980](#) for discussion and much empirical motivation.

In the strict conditional theory, by contrast, the negation of *If A, C* is much weaker than *If A, not C*. The latter is true just in case the conjunction $A \wedge C$ is impossible (true in no accessible worlds). By contrast, *not(if A, C)* asserts merely that $A \wedge \neg C$ is possible (i.e., true in *at least one* accessible world). As a result, Bert’s reply in (3b) is wrongly predicted to be consistent. This result is essentially due to the presence of a wide-scope universal quantifier in the definition of *if*.

There are various workarounds available: for instance, in the variably strict theory of [Stalnaker \(1968\)](#), commutation holds because the domain of quantification is required to be a singleton. A more recent suggestion is to make use of a homogeneity requirement to enforce commutation with negation ([Križ, 2015](#); [von Fintel, 1997](#)). These approaches may be workable, but they do not help with the next issue.

Desideratum 2. Probabilities of conditionals.

Intuitively, the probability of a conditional— $P(\textit{If } A, C)$ —is equal to the conditional probability $P(C \mid A)$. For instance, the probability of (4) is obviously $1/2$.¹

(4) If I flip this fair coin, it will come up heads.

[Stalnaker \(1970\)](#) argued that enforcing this identity is an adequacy criterion for a good theory of conditionals:

Stalnaker’s Thesis (ST). The correct semantic theory of conditionals will associate *If A, C* with an interpretation such that $P(\textit{If } A, C)$ is equal to $P(C \mid A)$.

Much subsequent experimental research provides evidence that ST is a correct claim about *if* ([Douven & Verbrugge, 2010](#); [Evans & Over, 2004](#); [Over & Evans, 2024](#)).

[Stalnaker \(1970\)](#) also claimed that the semantic theory proposed in [Stalnaker 1968](#) enforced this identity, and that this constituted an argument in its favor. However, David Lewis (1976) proved that Stalnaker’s semantic theory does not in fact enforce this equation, and indeed that no theory with a handful of seemingly innocuous properties could do so. We will revisit this issue below, because Lewis’ proof tacitly assumes bivalence and trivalent theories are able to circumvent it. For current purposes, though, the crucial observation is that strict conditional theories make plainly incorrect predictions about the probabilities of conditionals. Specifically, they predict that *If A, C* is false at every world in the domain if there are *any* worlds in the domain which which A is true and C false.²

As a result, if we are certain that $A \wedge \neg C$ is possible, then we can be certain that *If A, C* is false: it should have probability 0 ([Edgington, 1995, 2014](#)). This result is plainly incorrect. Consider (4): if the coin is fair and the speaker might flip it, then we can be certain that there is an epistemically accessible world in which the antecedent is true and the consequent false. Yet (4) has probability $1/2$, not 0.

Here again, the culprit is the universal quantifier. In strict conditional theories, as we transition from a set of accessible worlds with zero $A \wedge \neg C$ -worlds to a set with one or more such worlds, the probability of *If A, C* transitions abruptly from 1 to 0. The intuitively correct result, Stalnaker’s Thesis, would follow only if there were a way for the probability to decline gradually, in proportion to the number of $A \wedge C$ - vs. $A \wedge \neg C$ -worlds. Unfortunately, there does not seem to be any way for a semantics built on quantification over possible worlds to encode such a gradual decline in probability.

Desideratum 3. Update: Indicative = Material.

In C.I. Lewis’ work, a primary motivation for abandoning the material conditional analysis is that it is too weak: it makes too many conditionals true (notably, all false-antecedent conditionals) and generates too many validities (for instance, the argument in (1)). The success of modal theories in general, and the strict conditional theory in particular, has been due in no small part to the fact that they render the indicative logically stronger than the material conditional. In modalized strict conditional theories, this is because the truth of *if A, C* at w depends not just on what is happening at w , but also on other worlds accessible from w . For example, in the basic strict conditional theory in (2), $A \supset C$ can be true at

¹Notably, the probability of (4) does not seem to depend on how likely it is that the speaker will flip the coin, as long as it is possible that she will. Theories in which conditionals with false antecedents can be straightforwardly true or false will inevitably predict that worlds where the antecedent of (4) is false should matter to its probability.

²This gloss assumes that the epistemic accessibility relation is an equivalence relation; I take this to be uncontroversial.

w while *if* A , C is false, as long as $R(w)$ contains worlds other than w itself. Just let A and C be true at w , while there are other worlds in $R(w)$ where A is true and C false.

However, Khoo (2022) has recently pointed out that this creates a problem (his first “Bounding Puzzle”).³ If *if* is stronger than $\supset$, then a standard pragmatics of assertion (Stalnaker, 1978) predicts that update by *if* A , C should eliminate a strict superset of the possible worlds that are eliminated in update with $A \supset C$, at least in some contexts. As a result, it should make sense to assert $A \supset C$ while denying *if* A , C : the point of such a conversational move would be to locate the actual world in the intermediate logical space between the two.

Consider the following examples. Uncontroversially, it is incoherent to assert the indicative while denying the corresponding material conditional. This is what we would expect if *if* is stronger than $\supset$:

- (5) a. If Mary is not in Paris, she’s in Amsterdam.
- b. # But it’s not true that she’s either in Paris or in Amsterdam.

Surprisingly, though, the converse is also incoherent:

- (6) a. Mary is either in Paris or in Amsterdam.
- b. # But it’s not true that if she’s not in Paris, she’s in Amsterdam.

(6) suggests that an assertion of the material disjunction is inconsistent with the denial of the corresponding indicative conditional. It seems that a speaker who asserts $A \vee C$ —which is equivalent to $\neg A \supset C$ —is committed to the full informational content of *If* A , C , and vice versa. This fact is very puzzling, given that nearly everyone agrees that the two are not semantically equivalent, and most theorists would endorse the claim that *if* is logically stronger. As Khoo argues in detail, a theory of conditional semantics and pragmatics should be able to explain why (6) is incoherent.

Our third desideratum, then, is to explain the close connection between indicative conditionals and disjunction in a non-stipulative way. Strict conditional theories may in some cases be able to rise to the task, as Stalnaker (1975) and Khoo (2022) show. However, in both cases accounting for the data requires making substantive additional pragmatic assumptions, and it is not clear whether the solution extends to other strict conditional theories.

2.2 The restrictor theory

David Lewis (1975) points out a further problem for strict conditional theories, this time involving compositional interactions between the conditional antecedent and consequent.

- (7) a. If Caesar woke early, he always took a walk before breakfast.
- b. Often, if it is raining, the roof leaks.

Lewis argues that the only semantic function of the *if*-clauses in (7) is to restrict the quantificational domain of the adverbs of quantification *always* and *often*.

I conclude that the *if* of our restrictive *if*-clauses should not be regarded as a sentential connective. It has no meaning apart from the adverb it restricts. The *if* in *always if* ..., ..., *sometimes if* ..., ..., and the rest ... serves merely to mark an argument-place in a polyadic construction. (D. Lewis 1975: 11)

D. Lewis (1975) was content to postulate this as the function of *if* in a handful of constructions, and did not attempt to develop it into a general theory of conditionals or to reconcile it with the evidence for the (variably) strict interpretation that he advocated in D. Lewis 1973 and elsewhere. This more ambitious task was left to Kratzer (1991a), who argued that Lewis’ restrictor analysis applies to all uses of *if*:

The history of the conditional is the story of a syntactic mistake. There is no two-place *if* ... *then* connective in the logical forms for natural language. *If*-clauses are devices for restricting the domains of various operators. Whenever there is no explicit operator, we have to postulate one. (Kratzer 1991a: 656)

The crucial motivation for Kratzer’s approach is that examples analogous to (7) are also found among overt modals, including epistemic and deontic items among others. This gives us our fourth key phenomenon, which strict conditional theories do not account for, but the restrictor theory does.⁴

³Because of the restriction phenomena discussed below, this argument applies only to “bare” indicatives, i.e., those whose consequent does not contain an overt modal, adverb of quantification, or other overt operator whose domain is affected by the conditional antecedent.

⁴It is, however, possible to supplement a strict conditional theory by allowing the antecedent to have domain-shifting effects along the lines of the restrictor theory: see e.g. Gillies 2010.

Desideratum 4. Domain restriction of overt modals.

- (8) a. If Bill woke early today, he must have taken a walk.
b. If you've offended someone, you should apologize.

In these examples, the overt modals *must* and *should* are restricted to worlds in which the antecedent holds. (8a) is about what we can infer *on the assumption* that Bill woke early. Similarly, (8b) is about what you should do in cases where you've offended someone, and makes no claim about what you should do in other cases. The strict conditional analysis has no account of the data in (8): the $\Box$ in the interpretation of *if*-sentences is lexically associated with *if* and has no effect on the domains of other operators.

How, then, can we have a strict conditional analysis and also explain the domain-restricting effects of *if*-clauses? Kratzer's ingenious solution is to decompose the strict conditional analysis into two components: a restriction operation picked out by the *if*-clause, and a separate (overt or silent) modal operator. On a technical level, this is accomplished by postulating a bespoke parameter of interpretation g , the "modal base", which controls the domain of the relevant sentential operator. Simplifying a bit, we can think of g as a function taking a world w to the worlds $g(w)$ that are accessible from it.

$$(9) \quad \llbracket \text{if } A, \text{ MUST } C \rrbracket^{w,g} = \llbracket \text{MUST } C \rrbracket^{w,g_A}, \text{ where } g_A(w) = g(w) \cap \{w' \mid \llbracket A \rrbracket^{w',g} = 1\}.$$

As long as the silent *MUST* has the semantics of the $\Box$ in the strict conditional analysis, quantifying over epistemically accessible worlds, then Kratzer's analysis with silent *MUST* is semantically equivalent to the strict conditional analysis in (2). In the strict conditional theory, accessible worlds where A is false render the material conditional $A \supset C$ true, and so they do not falsify the universal modal. In Kratzer's variant, these worlds are ignored as being outside the domain of the modal. In either case, the net effect is to claim that there are no accessible $A \wedge \neg C$ -worlds.⁵

$$(10) \quad \llbracket \text{if } A, \text{ MUST } C \rrbracket^{w,g} = \llbracket \text{MUST } C \rrbracket^{w,g_A} = 1 \text{ iff, for all } w' \in g_A(w), \llbracket C \rrbracket^{w',g_A} = 1.$$

A major advantage of Kratzer's approach is that it is able to capture the phenomena motivating the (variably) strict conditional for bare conditionals, while effortlessly handling the numerous cases in which the *if*-clause restricts an overt modal, such as (8). All that is required is that the modals in (8) get their domains from the same g parameter that controls the domain of the silent *MUST* in (9). For a conditional with deontic *should*, this yields:

$$(11) \quad \llbracket \text{if } A, \text{ should } C \rrbracket^{w,g} = \llbracket \text{should } C \rrbracket^{w,g_A} = 1 \text{ iff } \forall w' \in \text{BEST}(g_A(w)), \llbracket C \rrbracket^{w',g_A} = 1.$$

Despite these attractions, Kratzer's theory inherits all of the drawbacks of the strict conditional theory that were discussed in the previous section, related to the presence of the universal quantifier. First, conditionals do not commute with negation, and negated conditionals have an excessively weak interpretation: negating *If Bill was hungry, he ate* yields an interpretation along the lines of *If Bill was hungry he might not have eaten* rather than the correct *If Bill was hungry, he did not eat*. For the same reason, Kratzer's theory makes implausible predictions about the probabilities of conditionals: (4) is predicted to have probability zero, since we can be certain that it is not true everywhere in the restricted domain.⁶ Third, the restrictor theory inherits the problem around Khoo's Bounding Puzzle: the update effect of an indicative conditional is stronger than that of a material conditional.

Beyond these inherited issues, Kratzer's restrictor theory introduces two new problems.

Desideratum 5. No silent operators.

Simply put, there is no independent evidence, syntactic or semantic, for the existence of a silent *MUST* that can appear in bare conditionals. Kratzer's sole argument for postulating one is that the decomposition of the strict-conditional theory that she is pursuing will not work without it. Presumably

⁵Kratzer's theory is, however, more flexible in two ways. First, she allows that silent operator may not be epistemic. This is motivated, for example, by bare conditionals with a generic or habitual interpretation, e.g. *If Sal is hungry, she eats pizza* (Farkas & Sugioka, 1983). Second, there is no requirement that the silent *MUST* have the interpretation of the $\Box$ of classical modal logic. Indeed, Kratzer (1981, 1991b) argues for an interpretation of overt *must* that is weaker than $\Box$: see von Fintel and Gillies 2010 for discussion and critique.

⁶Egré and Cozic (2011) suggest that Kratzer's predictions can be defended, if we assume that people systematically confuse questions about the probability of a conditional with questions about the probability (as restricted by the antecedent) of its consequent. See also Kratzer 2012: §4 for a similar claim. Methodological worries aside, this approach encounters problems in dealing with cross-speaker probability judgments with sentential anaphors, among other issues: see, e.g., Huitink 2008; von Fintel 2011.

methodological considerations should lead us to prefer a solution that does not require unmotivated covert operators if possible (cf. Ciardelli 2022; Schulz 2010; several options will be discussed below).

To be clear, there is reason to think that there is a silent *GEN* in bare conditionals with a generic interpretation. But the existence of these silent operators is motivated by the interpretation of non-conditional generics: the argument for a silent operator in *If Sal is hungry, she eats pizza* is exactly the same as the argument for one in *Sal eats pizza* (Farkas & Sugioka, 1983; Krifka et al., 1995). By contrast, there is no plausible argument for the presence of a silent epistemic *MUST* in non-conditional declarative sentences, even optionally.

Desideratum 6. Explaining systematic parallels between domain restriction and presupposition projection.

In Kratzer’s theory, the restriction operation targets a special-purpose modal domain parameter that controls the domains of sentential operators and does nothing else. However, there is reason to believe that the approach is insufficiently general: there are systematic parallels between modal domain restriction and presupposition projection phenomena. First, note that the way that conditional antecedents restrict operator domains in Kratzer’s theory precisely parallels the way that they restrict the local context in satisfaction-based theories of presupposition.

- (12) If Bill is a diver, he forgot his scuba gear.

In theories of presupposition following Heim (1983); Karttunen (1974); Stalnaker (1973), this sentence can be asserted felicitously in a context c only if every world in the context *in which Bill is a diver* is one in which he has scuba gear. In other words, the context in which the presupposition is evaluated is temporarily restricted by the content of the antecedent: when the consequent is evaluated, the global context c is replaced temporarily by the local context $c_A = c \cap \{w \mid \llbracket A \rrbracket^{c,w} = 1\}$. The requirement that (12) places on the global context c is thus that all worlds in c in which Bill is a diver are also worlds where Bill has scuba gear. As a result, (12) presupposes the material conditional *Bill is a diver* $\supset$ *Bill has scuba gear*.⁷

If we simply combine the satisfaction theory of presupposition with the restrictor theory of conditionals, this parallel turns out to be merely accidental: the conditional simultaneously restricts both the context parameter c and the domain parameter g in the same way. However, similar parallels arise in other contexts, making it unlikely that this is a coincidence. First, note that the domain of a modal in the second clause of a disjunction is restricted by the negation of the first disjunct.

- (13) Either Bill is at work, or he must be in bed already.

Generally, a disjunction is assertible only if the speaker considers both clauses possible. However, (13) is assertible by someone who is confident that *Bill must be in bed already* is false. Indeed, a speaker of (13) will *have to* be confident that the right disjunct is false in isolation, since an assertion of (13) implicates that Bill might be at work (and thus not in bed).

We can arrive at the correct interpretation of (13) by allowing *must* to be restricted by the negation of *Bill is at work*. (13) is thus equivalent to (14), with an overt conditional in the second disjunct.

- (14) Either Bill is at work, or—if he is not—he must be in bed already.

The same holds for deontic modals, for example:

- (15) a. Either Sam is studying, or he should be helping his parents in the yard.
b. $\approx$ Either Sam is studying, or—if he isn’t—he should be helping his parents in the yard.

Again, the interpretation is that Sam should be helping, on the assumption that he is not studying.

Since the restrictor theory is specifically tailored for conditionals, it has nothing to say about domain restriction by disjunction. This must be treated as a separate phenomenon. Now, one could deal with this issue by stipulation, modifying the lexical entry for disjunction so that the modal base parameter in the second disjunct is restricted by the negation of the first:

⁷In a bit more detail: broadly following Karttunen (1974), we assume that an assertion is pragmatically infelicitous unless each atomic constituent’s presupposition is entailed by its local context. That is, all sentences are parameterized by a context c , and, if p triggers presupposition $\underline{p}$, then p is a presupposition failure at c unless $\forall w' \in c : \llbracket \underline{p} \rrbracket^{c,w'} = 1$. Projection is governed by a simple pragmatic rule: a sentence is infelicitous if any its subconstituents is a presupposition failure relative to its local context. This is not the only implementation of the satisfaction theory available here, but it is the most straightforward and is sufficient for our purposes here.

$$(16) \quad \llbracket A \text{ or } B \rrbracket^{c,w,g} = 1 \text{ iff either } \llbracket A \rrbracket^{c,w,g} = 1 \text{ or } \llbracket B \rrbracket^{c,w,g_A} = 1.$$

However, this solution is inelegant because presupposition projection parallels domain restriction here as well (Klinedinst & Rothschild, 2012; Mandelkern, 2019).

$$(17) \quad \text{Either Bill isn't a diver, or he forgot to bring his scuba gear.}$$

Here too, the intuitive presupposition involves interpreting the second disjunct in a context restricted by the negation of the first: Bill has scuba gear in all worlds in c in which he *is* a diver, as in the explicitly conditionalized (18).

$$(18) \quad \text{Either Bill isn't a diver, or—if he is—he forgot to bring his scuba gear.}$$

To get the right result for the presuppositions of disjunctions, then, we would need to modify the definition of *or* from (16) to (19). This is inelegant: now we have two operators, *or* and *if*, each of which modifies the c and g parameters in exactly the same way.

$$(19) \quad \llbracket A \text{ or } B \rrbracket^{c,w,g} = 1 \text{ iff either } \llbracket A \rrbracket^{c,w,g} = 1 \text{ or } \llbracket B \rrbracket^{c_A,w,g_A} = 1.$$

The restrictor theory of conditionals and the satisfaction theory of presupposition projection have been developed largely in independence. However, close parallels between domain restriction and presupposition projection suggest that the correct theory should treat them as effects of a single change.

Table 1 gives a scorecard summarizing the discussion so far of the various theories and desiderata.

	MC	SC	VSCL	VSCS	RT
1. Commutes with negation	✗	✗	✗	✓	✗
2. Probability of $\Rightarrow$	✗	✗	✗	✗	✗
3. Update: $\Rightarrow = \text{MC}$	✓	✗	✗	(✓)	✗
4. Restriction	✗	(✓)	(✓)	(✓)	✓
5. No silent operators	✓	✓	✓	✓	✗
6. Restriction $\sim$ projection	✗	✗	✗	✗	✗

Table 1 Scorecard so far. Key: MC = material conditional; SC = strict conditional; VSCL = variably strict conditional à la Lewis; VSCS = Stalnaker conditional; RT = restrictor theory. The “(✓)” annotation is used in cases where the basic theory lacks a desirable property, but plausible extensions—possibly not discussed here—have it.

2.3 Restriction of local contexts

There is a straightforward solution to the problem just noted. If we combine the restrictor theory of conditionals and the satisfaction theory of presupposition naïvely, the result is inelegant: conditional antecedents and disjunctions separately restrict a modal base parameter and a context parameter in exactly the same way. The solution is to eliminate the bespoke modal domain parameter, and have both conditionals and disjunctions manipulate the parameter representing the local context. This parameter provides both the local contexts for presuppositions and the quantificational domains of (at least) epistemic and deontic modals. This is essentially the proposal of Klinedinst and Rothschild (2012); Yalcin (2007).⁸

$$(20) \quad \llbracket \text{if } \phi, \psi \rrbracket^{c,w} = 1 \text{ iff, for all } w' \in c_\phi, \llbracket \psi \rrbracket^{c_\phi,w'} = 1.$$

$$(21) \quad \llbracket \phi \text{ and } \psi \rrbracket^{c,w} = 1 \text{ iff } \llbracket \phi \rrbracket^{c,w} = 1 \text{ and } \llbracket \psi \rrbracket^{c_\phi,w} = 1.$$

$$(22) \quad \llbracket \phi \text{ or } \psi \rrbracket^{c,w} = 1 \text{ iff either } \llbracket \phi \rrbracket^{c,w} = 1 \text{ or } \llbracket \psi \rrbracket^{c_{\neg\phi},w} = 1.$$

$$(23) \quad \llbracket \text{not } \phi \rrbracket^{c,w} = 1 \text{ iff } \llbracket \phi \rrbracket^{c,w} = 0.$$

⁸See also Carter 2021 for a similar theory of the interpretation of *if*. Yalcin and Klinedinst & Rothschild’s official theories actually manipulate an “information state” parameter that is separate from the context. See their papers for a variety of motivations. Since these issues are not crucial for the discussion here, we simplify by using a single context parameter.

This proposal combines elements of the restrictor and strict conditional theories: universal quantification is built into the lexical semantics of *if*, and the antecedent manipulates a contextual parameter. However, the parameter that is modified does a number of further semantic and pragmatic jobs.⁹

As long as the context c represents a set of epistemically accessible worlds, this proposal gives us another path to a strict-conditional interpretation of bare conditionals. However, it is able to cope with domain restriction and the parallel with presupposition projection in both conditionals and disjunctions. The key is that the context parameter is modified in both the consequent of a conditional and the right disjunct of a disjunction. This immediately accounts for the presupposition filtering facts noted above, for both conditionals and disjunctions. Domain-restricting effects also follow, as long as the relevant modals inherit their domains from their local contexts.¹⁰ Illustrating with schematic interpretations for epistemic *must* and deontic *should*:

- (24) a. $\llbracket \text{must } \psi \rrbracket^{c,w} = 1$ iff, for all $w' \in c$, $\llbracket \psi \rrbracket^{c,w'} = 1$.
b. $\llbracket \text{If } \phi, \text{ must } \psi \rrbracket^{c,w} = 1$ iff, for all $w' \in c_\phi$: for all $w'' \in c_\phi$: $\llbracket \psi \rrbracket^{c_\phi,w''} = 1$.
- (25) a. $\llbracket \text{should } \psi \rrbracket^{c,w} = 1$ iff, for all $w' \in \text{BEST}(c)$, $\llbracket \psi \rrbracket^{c,w'} = 1$.
b. $\llbracket \text{If } \phi, \text{ should } \psi \rrbracket^{c,w} = 1$ iff, for all $w' \in c_\phi$: for all $w'' \in \text{BEST}(c_\phi)$: $\llbracket \psi \rrbracket^{c_\phi,w''} = 1$.

What is crucial here is that, in the (b)-examples, the (innermost layer of) quantification associated with the modals is restricted to worlds where in the modal's local context. As a result, the *if*-clause is always true in the worlds over which the modal quantifies. While this formulation follows [Klinedinst and Rothschild \(2012\)](#) in making use of (a variant of) [Yalcin's \(2007\)](#) domain semantics, we could equally well use a relational semantics, probabilistic semantics, or some other account as long as the modals' domains are sensitive to their local contexts in a similar way.

This theory—call it the “Restriction of Local Contexts” approach—accounts for the main arguments in favor of the restrictor theory, while avoiding the need for silent modals and explaining the parallel between domain restriction and presupposition projection. However, it also inherits other drawbacks of the restrictor theory associated with the universal quantification in the interpretation of *if*: it makes incorrect predictions about the probabilities of conditionals; *if* does not commute with negation; and it does not account in any obvious way for Khoo's Bounding Puzzle.

2.4 Trivalent truth-functional theories

Other than the material conditional, the theories discussed so far have all been variants on C.I. Lewis' (1912) strict implication analysis. As we saw in §2.1, Lewis' main argument for adopting an intensional analysis of the conditional was by process of elimination. The material conditional analysis makes demonstrably false predictions, especially about the truth-values and entailments associated with false-antecedent conditionals. Since the material conditional is the only bivalent truth-function that is even remotely plausible, the conclusion is that an intensional analysis is needed.

The reasoning is sound, as long as bivalence is taken for granted. However, a number of authors over the years have pointed out that a three-valued truth-functional analysis of *if* does not encounter the problems that motivated Lewis to abandon the material conditional. The basic trivalent semantics, originally proposed by [de Finetti \(1936\)](#), agrees with the material conditional analysis except that a

⁹A further question is whether the entry for disjunction should be “symmetric”, allowing material in the second disjunct to filter presuppositions of the first disjunct:

(1) $\llbracket \phi \text{ or } \psi \rrbracket^{c,w} = 1$ iff either $\llbracket \phi \rrbracket^{c,w} = 1$ or $\llbracket \psi \rrbracket^{c,w} = 1$.

There is considerable experimental evidence for symmetric filtering in disjunctions, while conjunction is strictly asymmetric as in (21). See [Kalomoiros and Schwarz 2024](#) for a survey and new experimental evidence. This point could affect the viability of a unified theory of presupposition projection and modal domain restriction, if it turns out that presupposition projection is symmetric but modal domain restriction is not. The question has not been systematically investigated in previous work, but see [McHugh 2024](#) for many relevant examples. In general, right-to-left filtering (2a) and restriction (2b) appear to be available in similar contexts.

(2) a. Either Bill forgot to bring his scuba gear, or he isn't a diver.
b. Either Sam should be helping his parents in the yard, or he's inside doing his homework.

If so, the robust parallel between projection and restriction pursued here is compatible with the existence of symmetric readings of disjunction. This issue would benefit from further empirical attention, however.

¹⁰As (24b) and (25b) make clear, this approach predicts that explicitly modalized conditionals have two layers of quantification: the one provided lexically by *if*, and the one associated with the overt modal. In the implementation discussed here, the outer layer of quantification is vacuous. Whether this is correct is debatable, but we will not pause to deal with this issue here as we will shortly be discarding the outer, lexical layer of quantification for other reasons.

conditional with a false antecedent is neither true nor false: it receives a third truth-value $\#$, which may be glossed “undefined” or “irrelevant”. As a result of this change, the “paradoxes of the material conditional” do not arise.

(26) **Basic trivalent semantics for *if***: “If ϕ , ψ ” is

- a. true (1) if ϕ and ψ are both true;
- b. false (0) if ϕ is true and ψ is false;
- c. undefined ($\#$) if ϕ is false.

Numerous subsequent authors have given arguments in favor of such an analysis, from linguistic, logical, probabilistic, and psychological grounds: see, among many others, [Baratgin, Over, and Politzer 2013](#); [Belnap 1970](#); [Cantwell 2008](#); [Cooper 1968](#); [Égré et al. 2021](#); [Lassiter 2020](#); [Milne 1997](#); [Over and Evans 2024](#); [Rothschild 2014](#); [Wason 1966](#).

To give a full semantic theory for a trivalent conditional, it is necessary to specify the handling of undefined antecedents and how the connectives treat the third truth-value. These questions have received much attention in recent years, especially in the careful work of [Égré, Rossi, and Sprenger](#) on the logic and probabilities of trivalent conditionals ([Égré et al. 2021](#) et seq.). We will return to these issues as they become relevant below.

The attractions of the trivalent theory are largely complementary to those of the Restriction of Local Contexts theory. Because the interpretation of the conditional does not involve quantification, it is well-placed to deal with Desiderata 1-3, which create trouble for strict conditional theories.

First, the trivalent conditional commutes with negation. The generally agreed entry for *not* in the trivalent semantics reverses true and false values, while preserving undefinedness.

- (27) $\llbracket \text{not } \phi \rrbracket^{c,w} =$
- a. 1 if $\llbracket \phi \rrbracket^{c,w} = 0$.
 - b. 0 if $\llbracket \phi \rrbracket^{c,w} = 1$.
 - c. otherwise, $= \#$.

Applying this rule to (26), we can see that *not(if ϕ , ψ)* takes $\phi \wedge \psi$ -worlds to 0, $\phi \wedge \neg\psi$ -worlds to 1, and $\neg\phi$ -worlds to $\#$. This is, of course, exactly the same as the interpretation of *If ϕ , $\neg\psi$* .¹¹

Second, as noted already by [de Finetti \(1936\)](#), the trivalent conditional easily enforces Stalnaker’s Thesis: the probability of *If A , C* is equal to the conditional probability of C given A . [Lassiter \(2020\)](#) points out that the famous triviality proof of [D. Lewis \(1976\)](#) tacitly assumes bivalence, as do a number of other triviality proofs for Stalnaker’s Thesis. With a third value in hand, we can define a probability measure P which enforces this equation by ignoring false-antecedent cases. Assuming a standard bivalent, finitely additive probability measure $prob : \mathcal{P}(W) \rightarrow [0, 1]$, define the probability of a sentence, as interpreted at context c , as the ratio of the probability of worlds in c where the sentence is true to the probability of the worlds in c where it is defined.

$$P_c(\ulcorner \chi \urcorner) = \frac{prob(\{w \in c \mid \llbracket \chi \rrbracket^{c,w} = 1\})}{prob(\{w \in c \mid \llbracket \chi \rrbracket^{c,w} \in \{0, 1\}\})} \quad (28)$$

This definition reduces to the classical definition of probability whenever χ is bivalent (because then its denominator is always 1). When χ is a conditional *If ϕ , ψ* , though, it outputs the conditional probability of ψ given ϕ (as long as these are bivalent), interpreted using the [Kolmogorov 1933](#) ratio definition of conditional probability, $P(C \mid A) = P(C \wedge A)/P(A)$.

$$P_c(\ulcorner \text{If } \phi, \psi \urcorner) = \frac{prob(\{w \in c \mid \llbracket \text{If } \phi, \psi \rrbracket^{c,w} = 1\})}{prob(\{w \in c \mid \llbracket \text{If } \phi, \psi \rrbracket^{c,w} \in \{0, 1\}\})} = \frac{prob(\{w \in c \mid \llbracket \phi \wedge \psi \rrbracket^{c,w} = 1\})}{prob(\{w \in c \mid \llbracket \phi \rrbracket^{c,w} = 1\})} \quad (29)$$

Third, the trivalent semantics readily explains Khoo’s Bounding Puzzle (Desideratum 3). In [Stalnaker’s \(1978\)](#) influential theory of assertion, the conversational function of an assertion of χ is to propose

¹¹In the trivalent semantics both $A \Rightarrow C$ and $A \Rightarrow \neg C$ are undefined when A is false, but it is impossible for both sentences to be false at the same world. As a result, the Conditional Excluded Middle condition $A \Rightarrow C \vee A \Rightarrow \neg C$ is either true or undefined everywhere on any plausible trivalent definition of $\Rightarrow$ (including the definition given in §3.1 below). This means that CEM is logically valid according to the definition of validity as preservation of nonfalsity that [Égré et al. \(2021\)](#) argue is most appropriate for the trivalent semantics.

modifying the context c to a new context c' such that χ is true everywhere in c' . Equivalently, in Stalnaker's original account, an assertion of χ is a proposal to modify c so that χ is not false anywhere in c .

- (30) a. **Assertion, V1:** $ASSERT(\ulcorner \chi \urcorner, c) = c \cap \{w \mid \mathcal{I}(\ulcorner \chi \urcorner, c, w) = 1\}$.
b. **Assertion, V2:** $ASSERT(\ulcorner \chi \urcorner, c) = c - \{w \mid \mathcal{I}(\ulcorner \chi \urcorner, c, w) = 0\}$.

In the trivalent semantics, these two operations are not equivalent. If we restrict c to worlds where a conditional *If* ϕ , ψ is true as in V1, we end up with a new context containing only worlds where ϕ and ψ are both true. This is equivalent to updating with a conjunction—clearly the wrong result. By contrast, V2 gives a sensible result: an assertion of *If* ϕ , ψ is a proposal to remove worlds that make the conditional false, i.e., those where ϕ is true and ψ is false. Note, though, that these are the same worlds that make $\phi \supset \psi$ false. As a result, the only plausible extension of Stalnaker's theory of assertion for the trivalent semantics immediately accounts for Khoo's Bounding Puzzle: update with a (bare) trivalent conditional is equivalent to update with the corresponding material conditional, even though the two are not semantically equivalent.

A further attraction of the trivalent semantics is that it does not require postulating silent operators (Desideratum 5).

The basic trivalent semantics does not immediately account for the use of conditional antecedents to restrict operator domains (Desideratum 4). However, there are plausible extensions of the theory that do so. The basic idea is to define the lexical entries of modals (and other operators) so that their quantification is restricted to worlds in which the modal's prejacent has a value in $\{0, 1\}$. For example, *If Bill woke early, he must have taken a walk* checks that all accessible worlds w' in which $\llbracket \text{Bill woke early} \rrbracket^{c, w'} \in \{0, 1\}$, $\llbracket \text{Bill took a walk} \rrbracket^{c, w'} = 1$. This is the same as requiring that the non-modalized conditional *If Bill woke early, he took a walk* is either undefined or true in every accessible world. The lexical entry of the modal bakes in a particular pattern of sensitivity to the $\#$ value: whenever it appears, it is ignored. This idea is pursued by Huitink (2008), drawing inspiration from Belnap (1970). More recently, Égré, Rossi, and Sprenger (2025c) propose a theory that achieves a similar effect by defining modals as quasi-dynamic tests over local contexts that are shifted by the antecedent (also taking inspiration from the theory of Klinedinst and Rothschild 2012; Yalcin 2007). In both cases, the effect is that operators so defined will automatically be restricted to quantify over worlds that verify the antecedent.

However, this approach has certain drawbacks. First, domain restriction is a very general property of modal/conditional interactions, and does not seem to be an item-by-item eccentricity of particular expressions that would naturally be placed in the lexicon. For example, there is nothing in Huitink's trivalent theory of restriction that would prevent us from defining a modal *must'*, a variant of *must* that requires its domain to consist entirely of worlds where the conditional is true. This would make *If* ϕ , *must'* ψ equivalent to *must*($\phi \wedge \psi$). However, it seems doubtful that natural languages would allow such a modal to exist, either alone or alongside the attested *must*.¹²

The trivalent theory of restriction also shares several drawbacks of Kratzer's (1991a) original restrictor theory: it is focused too narrowly on conditionals, and does not account for the parallel use of disjunction to restrict operator domains. Nor does it provide an explanation of parallels between domain restriction and presupposition projection, notably domain restriction by disjunction.

We now have the updated scorecard in Table 2.

The explanatory challenge.

This challenge is specific to trivalent theories, which have been developed mainly by philosophers and logicians in response to technical issues. When we try to incorporate trivalent conditionals into a comprehensive theory of natural language semantics and pragmatics, certain conceptual puzzles arise. In the theory as developed so far, the undefined value appears only in sentences that contain *if*-clauses. But why would *if*-clauses be special in this way? Does it really make sense, in terms of the broader architecture of the semantic theory, to add a third semantic status merely to account for the quirks of a single construction type?

A natural response to this question is to note that, on many accounts, a third truth-value is already used to account for presupposition failure (Beaver and Krahmer 2001; George 2014; Peters 1979; Winter

¹²This criticism does not apply to the more recent proposal of Égré et al. (2025c), in which the modal domain is indexed to a context parameter that is shifted by *if*-clauses, as in Yalcin 2007 and the temporary-context theory outlined below.

	MC	SC	VSCL	VSCS	RT	RLC	3V
1. Commutes with negation	✗	✗	✗	✓	✗	✗	✓
2. Probability of $\Rightarrow$	✗	✗	✗	✗	✗	✗	✓
3. Update: $\Rightarrow$ = MC	✓	✗	✗	(✓)	✗	✗	✓
4. Restriction	✗	(✓)	(✓)	(✓)	✓	✓	(✓)
5. No silent operators	✓	✓	✓	✓	✗	✓	✓
6. Restriction $\sim$ projection	✗	✗	✗	✗	✗	✓	✗

Table 2 Updated scorecard. Key: MC = material conditional; SC = strict conditional; VSCL = variably strict conditional à la Lewis; VSCS = Stalnaker conditional; RT = restrictor theory; RLC = restriction of local context à la [Klinedinst and Rothschild 2012](#); [Yalcin 2007](#); 3V = trivalent conditional.

2025 a.m.o.). It might seem, then, that the use of a third value in trivalent theories of conditionals is without theoretical cost. This is not the case, however. Many theories of presupposition—including the account of [Karttunen \(1973, 1974\)](#) that I adapt below—are fully pragmatic, with failure resulting in infelicity rather than semantic undefinedness. Even if we assume that some trivalent theory of presupposition is correct, combining this with a trivalent semantics for conditionals is highly non-trivial. The resulting account is fairly complex and requires modifying the trivalent semantics for conditionals in potentially problematic ways.¹³ Another possibility is that there are two orthogonal kinds of semantic “undefinedness”, one associated with presupposition failure and the other with false-antecedent conditionals. In that case, however, the use of a third value does little conceptually to motivate a fourth.

A further aspect of the explanatory challenge for trivalent semantics of conditionals is to explain why false-antecedent conditionals are systematically treated as “ignorable” or “irrelevant”, for example, when computing probabilities or in calculating the domains of modals. If $\#$ is a truth-value on a par with 1 and 0, there is no formal reason why probability calculations should exclude this value rather than collapsing it with one of the other values, or choosing to attend to $\#$ -values while excluding one of the other values. The same observation goes for modal domains. A trivalent theory could, in principle, allow modals to specify on an item-by-item basis how the $\#$ value should be handled. In fact, though, all modals handle this value in the same way: false-antecedent worlds are excluded from consideration in checking whether the quantificational requirements of a modal are satisfied. More broadly, the trivalent theory, even if it is fully adequate as a logic, gives little insight into why the theory is structured in precisely the way that it is, and how this relates to broader themes in natural language interpretation.

A different response to the first part of the challenge is to point out that the third value may not be needed only for *if*-clauses: very plausibly, it is needed to account for other constructions with a suppositional interpretation (such as *supposing*, *assuming*, *on the assumption that*, *in case*, *otherwise*, *provided*, *in case*, *barring*, as well as modal subordination and more). However, I will argue that the issue goes still deeper. Inspired in particular by [Schlenker \(2009\)](#), the theory proposed below proceeds on the assumption that we strive in natural language interpretation to restrict our attention as far as possible. Local contexts thus function as a way of keeping track of which possibilities we need to attend to, and which we can safely ignore. The semantic and pragmatic apparatus sketched follows this intuition: worlds outside the local context are not treated as true or false, but simply as irrelevant. On this account, the trivalent status of indicative conditionals is not a stipulated feature of *if* or any other operator, but emerges as a predictable side effect of the way that these items manipulate local contexts.¹⁴

¹³In brief, a successful account along these lines requires formally differentiating the effect of presupposition failure from that of a false-antecedent conditional: after all, conditionals do not presuppose the truth of their antecedents. In addition, conjunctions of conditionals behave differently from conjunctions of presuppositional expressions. For the latter, the treatment of conjunction associated with the Cooper-Cantwell semantics discussed below wrongly predicts that examples such as *Mary is glad that it's raining*, and *Bill is glad that it's not raining* should express a coherent thought.

In recent work [Booth \(2025\)](#) proposes a different way to combine three-valued presuppositions with three-valued conditionals within a dynamic pragmatic theory. However, Booth's account essentially gives up on the explanatory project because it requires stipulatively associating distinct pragmatic effects with conditional and non-conditional sentences.

¹⁴See §6.3 for further discussion of Schlenker's theory of local contexts and the present theory's relation to it, particularly with respect to Schlenker's problem of how to render a theory of local contexts explanatory.

3 The temporary-context theory

The temporary-context theory addresses the explanatory challenge just sketched, while maintaining the positive features of the trivalent account and combining it with features of the theory based on restricted local contexts. In slogan form, the theory can be summarized in two brief claims, only one of which is specific to conditionals:

- In natural language interpretation, worlds outside the local context are systematically treated as **irrelevant**.
- *If*-clauses are instructions to create a **temporary context**.

The rest of the theory emerges in interaction with other pieces, most of which are independently motivated.

In its technical implementation, the temporary-context theory is modeled closely on the approach of Yalcin (2007) and Klinedinst and Rothschild (2012) discussed in §2.3, with two crucial differences. First, the conditional is not modal or quantificational: it simply uses the antecedent to temporarily restrict the input context. Second, the denotation function $\llbracket \cdot \rrbracket^{c,w}$ outputs an undefined value $\#$ for any world w that lies outside the local context c .

The main idea behind the temporary-context theory—“*If*-clauses restrict the context and do nothing else”—is not new. The motivating intuition is close to conditional assertion theories from Ducrot 1972; Mackie 1973 to Stalnaker 2011, though these are usually not formalized. The immediate inspiration for the present version is the dynamic semantics of Kaufmann (2000), who uses it to treat modal subordination and derives the equivalence of update with an indicative conditional and a material conditional. Kaufmann’s paper and work following it have largely focused on discourse-level phenomena such as modal subordination (Kaufmann, 2000), conditional questions (Isaacs & Rawlins, 2008), *what if?* questions (Bledin & Rawlins, 2019), and bare conditional antecedents in Spanish (Biezma, 2025). Here we take a complementary line, focusing on sentence-level compositional phenomena that have been the bread and butter of most semantic work on conditionals.¹⁵

3.1 Semantics

For simplicity, we assume a propositional language with variables $p, q, r, \dots$ over atomic sentences, and variables $\phi, \psi, \dots, A, B, \dots$ over possibly complex sentences generated by the grammar

$$\phi := p \mid \phi \vee \phi \mid \phi \wedge \phi \mid \phi \Rightarrow \phi \mid \Box_e \phi \mid \Box_d \phi.$$

As above, we often use English equivalents for readability: so “if ϕ, ψ ” is interchangeable with “ $\phi \Rightarrow \psi$ ”, “ $\Box_e \phi$ ” and “ $\Box_d \psi$ ” with “ $\phi \wedge \psi$ ”, and so on.

$c, c', \dots$, are variables over contexts, treated as sets of possible worlds. We draw a crucial distinction between the interpretation function $\mathcal{I}$, which can be applied to arbitrary sentences of the language, and the denotation function $\llbracket \cdot \rrbracket^{c,w}$ which assigns truth-values to atomic sentences only (relative to a context and a possible world).

The denotation function $\llbracket \cdot \rrbracket^{c,w}$ is a function mapping an atomic sentence, a context, and a world to $\{0, 1, \#\}$. Our crucial assumption is that the denotation of an atomic sentence, relative to a context c and a world w , is undefined ($\#$) if, and only if, w is not in the context c . Otherwise $\llbracket p \rrbracket^{c,w}$ maps p to its (bivalent) truth-value in w . Essentially, atomic sentences denote binary classifiers restricted to the context in which they occur, and treat any world outside the local context as semantically irrelevant.

¹⁵Compare also the proposal of Starr (2014), which is very close to Kaufmann’s (2000). However, Kaufmann’s theory treats indicatives as conveying the same information as a material conditional—correctly, as we argued above—while Starr analyzes them as dynamic tests that do not convey new information. Starr’s theory also resembles de Finetti’s semantics in that false-antecedent conditionals are not assigned truth-values, but this status is derivative and does not do compositional work as it does in the present account (Starr 2014: 14, 22).

Another variant of Kaufmann’s (2000) view is developed by Ciardelli (2021). This view is also similar to the temporary-context theory in that it starts with a static semantics based on Klinedinst and Rothschild 2012; Yalcin 2007. Ciardelli’s proposal depends on the assumption that *if*-clauses can restrict either the domain of an overt operator or an information state parameter, and that in the latter case conditionals (and other modal sentences) have expressive rather than truth-conditional meanings. While a detailed comparison is not possible in this space, it does seem that the temporary-context theory is at a simplicity advantage in treating *if*-clauses as having the same semantic effect everywhere, and in assigning a straightforward factual interpretation to conditional sentences. The latter also helps to explain mutual entailments between conditional and non-conditional (e.g., disjunctive) sentences that are less straightforward if one is truth-conditional and the other expressive.

The interpretation function $\mathcal{I}$ is defined recursively as follows. The interpretation of an atomic sentence $\ulcorner p \urcorner$ at (c, w) is simply its truth-value at (c, w) .

$$(31) \quad \mathcal{I}(\ulcorner p \urcorner, c, w) = \llbracket p \rrbracket^{c,w}$$

To define *if* and the connectives, it will be helpful to define an abbreviation for a restricted context.¹⁶

$$(32) \quad c_\phi \text{ is the \textbf{restriction} of } c \text{ to the worlds where } \phi \text{ is not false: } c_\phi =_{df} c \cap \{w' \mid \mathcal{I}(\ulcorner \phi \urcorner, c, w') \neq 0\}.$$

The interpretation of the indicative conditional is simply:

$$(33) \quad \mathcal{I}(\ulcorner \text{if } \phi, \psi \urcorner, c, w) = \mathcal{I}(\ulcorner \psi \urcorner, c_\phi, w)$$

In words: *If* ϕ , ψ at context c and world w is equivalent to ψ at the same world, relative to a temporary context c_ϕ which is restricted by the assumption that ϕ holds.¹⁷

It follows from this definition, and the definition of the restriction operation above, that indicative conditionals are trivalent. In the simplest case, and assuming that the world of evaluation is in the context, a conditional *if* p , q with atomic p and q is true at w when p and q are both true at w ; false when p is true and q is false at w , and undefined when p is false at w . This pattern follows from the interaction of local contexts and the fact that $\llbracket \cdot \rrbracket^{c,w}$ outputs a truth-value only for worlds w in the input context c .

$$(34) \quad \mathcal{I}(\ulcorner \text{if } p, q \urcorner, c, w) = \mathcal{I}(\ulcorner q \urcorner, c_p, w)$$

- a. is $\#$ when p is false at w or $w \notin c$. This is because, whenever $\mathcal{I}(\ulcorner p \urcorner, c, w) = 0$, the restriction to c_p will **exclude** the evaluation world w . As a result, $\mathcal{I}(\ulcorner q \urcorner, c_p, w) = \#$ by the definition of $\llbracket \cdot \rrbracket^{c,w}$.
- b. has the truth-value of q whenever $w \in c$ and p is true at w . This is because, whenever $\mathcal{I}(\ulcorner p \urcorner, c, w) = 1$, the restriction to c_p will **include** the evaluation world w as long as $w \in c$. As a result, $\mathcal{I}(\ulcorner q \urcorner, c_p, w) = \llbracket q \rrbracket^{c,w}$ by the definition of $\llbracket \cdot \rrbracket^{c,w}$.

More generally, *if* ϕ , ψ is undefined at (c, w) when ϕ is false at (c, w) , for arbitrary ϕ . This is because *if* ϕ restricts c by eliminating worlds $w' \in c$ where ϕ is false at (c, w') , in the process excluding the evaluation world. In such cases, the interpretation function will always ground out in atomic denotations where the relevant world lies outside the relevant context, thus outputting $\#$. If ϕ is true at w , the conditional has the truth-value of its consequent. This is because the restriction operation does not eliminate non-false worlds from the context, and so the final result may be non- $\#$. Under the definition of the restriction operation given above, the same is true for undefined antecedents (see fn. 16).

Negation reverses truth and falsity, but leaves $\#$ -values unchanged.

$$(35) \quad \begin{aligned} \mathcal{I}(\ulcorner \text{not } \phi \urcorner, c, w) = \\ \# \text{ if } \mathcal{I}(\ulcorner \phi \urcorner, c, w) = \#; \\ 1 \text{ if } \mathcal{I}(\ulcorner \phi \urcorner, c, w) = 0; \\ 0 \text{ if } \mathcal{I}(\ulcorner \phi \urcorner, c, w) = 1. \end{aligned}$$

For conjunction and disjunction, we follow the proposal of Cooper (1968), which is given detailed motivation on both logical and probabilistic grounds by Égré et al. (2021, 2025a, 2025b).¹⁸ These definitions reduce to ordinary Boolean conjunction and disjunction whenever $w \in c$ and ϕ and ψ do not contain conditionals (or other $\#$ -inducing operators, in an expanded fragment).

¹⁶In combination with the other machinery in this section, this restriction operation allows the temporary-context theory to simulate the Cooper-Cantwell variant of the trivalent semantics, where undefined antecedents are handled like true antecedents. The obvious alternative definition is $c_\phi =_{df} c \cap \{w' \mid \mathcal{I}(\ulcorner \phi \urcorner, c, w') = 1\}$. This definition would simulate de Finetti's original semantics in terms of its handling of undefined antecedents. See Égré et al. 2021; Égré, Rossi, and Sprenger 2025a, 2025b for detailed discussion of the Cooper-Cantwell semantics and empirical and theoretical arguments in its favor.

¹⁷Given the definition in (32), “holds” must be interpreted in terms of non-falsity when ϕ is itself a conditional.

¹⁸The behavior of the $\#$ -value in these definitions is conceptually motivated within the broader temporary context theory, as discussed in the main text. Égré et al. (2021) discuss a number of further logical and probabilistic advantages of defining conjunction and disjunction in this way. The most straightforward argument begins with an objection to de Finetti's original proposal due to Bradley (2002). De Finetti makes use of the strong Kleene connectives, where $\# \wedge 1 = 1 \wedge \# = \#$. Bradley notes that, on this theory, the following example should be undefined everywhere, since one or the other of the conjuncts is always undefined.

(1) If it rains, we'll stay home; and if it doesn't rain, we'll go out.

By contrast, the definition in (36) gives the intuitively correct interpretation. If it rains, (1) is true or false depending on whether we go out, and likewise if it does not rain. In either case, the truth-value of the conjunction is determined by whichever conjunct is not being ignored.

- (36) $\mathcal{I}(\ulcorner \phi \text{ and } \psi \urcorner, c, w) =$
- a. 0 if $\mathcal{I}(\ulcorner \phi \urcorner, c, w) = 0$ or $\mathcal{I}(\ulcorner \psi \urcorner, c, w) = 0$;
 - b. # if $\mathcal{I}(\ulcorner \phi \urcorner, c, w) = \#$ and $\mathcal{I}(\ulcorner \psi \urcorner, c, w) = \#$;
 - c. otherwise, = 1.

In terms of our intuitive gloss of # as “ignorable”, this definition can be explicated as follows. If both conjuncts are true, the conjunction is true. If either conjunct is false, this is enough to conclude that the conjunction is false, without worrying about the truth-value of the other. If one conjunct is true and the other is #, then the conjunction is true because the #-value is treated as if it were not present. If both conjuncts are ignorable, then the entire conjunction is labeled as ignorable.

Disjunction has a similar interpretation. If either disjunct is true, this is enough to conclude that the disjunction is true, without worrying about the truth-value of the other. If one is false and the other is #, then the # value is ignored and the disjunction is false. If both disjuncts are ignorable, then the entire disjunction is ignorable.

- (37) $\mathcal{I}(\ulcorner \phi \text{ or } \psi \urcorner, c, w) =$
- a. 1 if $\mathcal{I}(\ulcorner \phi \urcorner, c, w) = 1$ or $\mathcal{I}(\ulcorner \psi \urcorner, c, w) = 1$;
 - b. # if $\mathcal{I}(\ulcorner \phi \urcorner, c, w) = \#$ and $\mathcal{I}(\ulcorner \psi \urcorner, c, w) = \#$;
 - c. otherwise, = 0.

The other key feature of *and* and *or* for current purposes is that they manipulate the context parameter, as in the Restriction of Local Context theory discussed above. For *and*, the local context of the second conjunct is restricted to worlds that make the first conjunct true. For *or*, the local context of the second disjunct is restricted by the negation of the first. (As noted in Fn. 9, a symmetric variant of *or* is also available, but we will not pursue this issue in detail here.)

Finally, we come to the epistemic and deontic modals. As we will see, our treatment of modals has two basic requirements. First, strong modals should be treated as generalized conjunction based on the definition of *and* in (36). Second, modal domains must be constrained by the local context in which the modals occur. These ideas are compatible with numerous theories of these modal operators, including the domain semantics of [Klinedinst and Rothschild \(2012\)](#); [Yalcin \(2007\)](#) discussed above and a relational (/broadly Kratzerian) semantics, among others. Rather than choosing a semantics for modals as part of the core theory, I will present several options in §4.5 below and show how they predict modal domain restriction as an automatic side effect of the context shift associated with *if*-clauses. The theory is, however, intended to be modular with respect to the correct theory of modal semantics.

3.2 Pragmatics

As with trivalent theories generally, an assertion of ϕ in context c is treated as a proposal to modify the context by removing worlds where ϕ is false.

- (38) **Assertion:** $ASSERT(\phi, c) = c - \{w \mid \mathcal{I}(\ulcorner \phi \urcorner, c, w) = 0\}$.

We will also make use of a pragmatic norm of assertion essentially derived from [Stalnaker \(1978\)](#):

- (39) **Don’t make trivial assertions:** a speaker S at context c should not assert ϕ unless S believes that ϕ is true at some worlds in c .

This condition has an obvious Gricean motivation ([Grice, 1975](#)): violating it would mean saying something that could not possibly be true, contrary to the maxim of Quality.

In the context of a trivalent theory of conditionals, including the temporary-context theory, the norm of assertion in (39) predicts that an assertion of *If* ϕ , ψ should be infelicitous unless there are worlds in c where ϕ is true. If ϕ were true nowhere in c , then *If* ϕ , ψ would be undefined everywhere in c , and so trivial. The trivalent theory thus derives, on independently motivated pragmatic grounds, the “presupposition of antecedent possibility” that [Stalnaker \(1975\)](#) and many following authors have found it necessary to stipulate for indicative conditionals (e.g., [Leahy 2011, 2018](#); [Schlenker 2008](#); [von Stechow 1998](#)).¹⁹

¹⁹[Santorio \(2025\)](#) independently makes a similar observation, and provides evidence from projection in quantified conditionals that supports the claim that antecedent compatibility is not a presupposition. If it were, then *Every student got an A if she worked hard* would presuppose *Every student might have worked hard*, contrary to fact. This is because, quite generally, presuppositions project universally from the scope of a universal quantifier (e.g., [Chemla 2009](#)).

The most straightforward approach to presupposition in the temporary-context theory is a pragmatic one inspired by Karttunen (1974). In this account, an assertion is pragmatically infelicitous unless each atomic constituent’s presupposition is entailed by its local context.

- (40) **Presupposition failure:** If p triggers presupposition $\underline{p}$, then p is a presupposition failure at c unless $\forall w' \in c : \llbracket \underline{p} \rrbracket^{c, w'} = 1$.
- (41) **Presupposition projection:** ϕ is infelicitous in c if any subconstituent ϕ' of ϕ is a presupposition failure, relative the local context in which ϕ' occurs.

It is possible to combine the temporary-context theory with a trivalent theory of presupposition (Beaver & Krahmer, 2001; George, 2014; Peters, 1979; Winter, 2025), but a number of formal and empirical issues arise with regard in the treatment of complex sentences. To avoid these technical issues, we prefer to treat presupposition failure in terms of pragmatic infelicity rather than semantic undefinedness.

4 Desiderata

By combining features of the theory based on local contexts and the trivalent semantics, the temporary-context theory fulfills all of the desiderata that were discussed above—uniquely among the theories discussed here. It also responds to the explanatory challenge for trivalent theories.

4.1 Commutation with negation

The temporary-context theory inherits much of its logic from the Cooper-Cantwell semantics (Égré et al., 2021, 2025a, 2025b). As one would expect, then, its conditional commutes with negation. Recall that this property creates trouble for (variably) strict conditional theories—except the Stalnaker conditional—and for their descendants including the restrictor theory and the modalized local context theory. Applying the definition of negation in (35), we get:

- (42) $\mathcal{I}(\ulcorner \text{if } \phi, \text{ not } \psi^\neg, c, w) =$
 - a. $\#$ if $w \notin c_\phi$; otherwise,
 - b. 1 if $\mathcal{I}(\ulcorner \psi^\neg, c_\phi, w) = 0$;
 - c. 0 if $\mathcal{I}(\ulcorner \psi^\neg, c_\phi, w) = 1$.
- (43) $\mathcal{I}(\ulcorner \text{not } (\text{if } \phi, \psi)^\neg, c, w) =$
 - a. $\#$ if $\mathcal{I}(\ulcorner \text{if } \phi, \psi^\neg, c, w) = \#$, i.e., if $w \notin c_\phi$;
 - b. 1 if $\mathcal{I}(\ulcorner \text{if } \phi, \psi^\neg, c, w) = 0$, i.e., if $\mathcal{I}(\ulcorner \psi^\neg, c_\phi, w) = 0$.
 - c. 0 if $\mathcal{I}(\ulcorner \text{if } \phi, \psi^\neg, c, w) = 1$, i.e., if $\mathcal{I}(\ulcorner \psi^\neg, c_\phi, w) = 1$.

4.2 Generalized Stalnaker’s Thesis

As we saw above, trivalent theories capture Stalnaker’s Thesis by stipulating that worlds where a sentence has the $\#$ value are ignored in the calculation of probabilities. The result is that the probability of a sentence is the probability that it is true, normalized by (i.e., divided by) the probability that it is semantically defined (true or false). If the sentence happens to be a conditional, then worlds where the antecedent is false do not affect the normalization. This ensures that the probability of the conditional is the conditional probability.

In the temporary-context theory, we can capture the same effect by thinking about the contexts in which probability claims are evaluated. For epistemic notions in general (cf. modal domain restriction, §4.5 below), only worlds in the context are taken into account, and those outside the context are ignored. If the context parameter for the probability assessment includes the information that the antecedent is true, then the probability of a conditional *If* ϕ , ψ in global context c is evaluated as the probability of ψ in c_ϕ , which turns out to be the same as the conditional probability of ψ given ϕ in c . This does not quite come for free, though: first we have to show that probability assessments are sensitive to manipulations of context in the same way as other aspects of semantic and pragmatic interpretation.

To start, we assume a bivalent probability measure $\text{prob} : \mathcal{P}(W) \rightarrow [0, 1]$. In the basic case, where we are evaluating a simple sentence against our actual information, this is straightforward: the probability

of a sentence, as interpreted at context c , as the ratio of the probability of worlds in c where the sentence is true to the probability of c itself.²⁰

$$P_c(\chi) = \frac{\text{prob}(\{w \in c \mid \mathcal{I}(\ulcorner \chi \urcorner, c, w) = 1\})}{\text{prob}(c)} \quad (44)$$

Conceptually speaking, it could have turned out that we always evaluate probabilities relative to a measure that is fixed by the actual informational context in which we actually find ourselves: say, our actual evidence, or some function of the evidence of the various people involved when there are multiple assessors. But probability assessments are in fact sensitive to the same factors that influence the context-sensitive interpretation of pronouns, modals, and other such expressions. The latter point has been studied in much detail in the literature on modal subordination (e.g., Roberts 1989; Stone 1999). In many cases we interpret context-sensitive expressions relative to a body of information that is built up in the discourse, and kept separate from the context of assessment—what Stalnaker (2014: §4) calls a “parallel context”. Processing the brief discourse in (45) (from Roberts 1989) involves building up a parallel discourse context in which Edna does in fact forget to fill the birdfeeder, she feels bad, and the birds get hungry.

- (45) a. If Edna forgets to fill the birdfeeder, she will feel very bad.
b. The birds will get hungry.

As discussed in detail in the modal subordination literature, all of this information is available in subsequent discourse to license anaphors and determine modal domains (46a), among other tasks. The same holds for probability evaluations (46b), which are implicitly conditioned on the same parallel context.

- (46) a. They might even leave permanently.
b. What is the probability that the birds will starve?

Similar considerations hold of parallel contexts in discussions of fiction: for example, I can wonder about the probability that the Gambler in the eponymous Kenny Rogers song won the last game of poker he played before his death, taking into account the information available in the song (e.g., he was highly experienced, but the song suggests that he was poor)—knowing full well that no such person ever lived.

When a statement or question about probabilities is assessed, then, the result is conditioned on the information in the relevant discursive context, rather than the information in the primary context. As a result, when we evaluate *If* ϕ , ψ in context c , the temporary context set up by the conditional antecedent *If* ϕ also affects the context parameter relevant to probability assessment: in effect, it creates a very small modal subordination context. On this assumption, the probability of an *if*-sentence should be:

$$P_c(\text{If } \phi, \psi) = P_{c_\phi}(\psi) = \frac{\text{prob}(\{w \in c_\phi \mid \mathcal{I}(\ulcorner \psi \urcorner, c_\phi, w) = 1\})}{\text{prob}(c_\phi)} \quad (47)$$

Since both the top and bottom of the fraction are restricted to ϕ -worlds, the probability of the conditional is the probability of $\phi \wedge \psi$ divided by the probability of ϕ , all relative to c . This is just the conditional probability in c of ψ given ϕ .²¹

Rather than stipulating that the denominator includes only worlds where the sentence is either true or false, as the standard trivalent theory does, the temporary-context theory captures Stalnaker’s Thesis with a very general assumption: epistemic operations, including probabilities, quite generally take into account only information in the relevant context. As we will see shortly in §4.5, this point connects directly to the explanation of modal domain restriction in the temporary-context theory.

In addition, the temporary-context theory predicts a slight generalization of Stalnaker’s Thesis. Since *if*-clauses are not the only way to modify the context parameter, an analogue of Stalnaker’s Thesis should hold whenever an expression introduces a local context that differs from the global context. This appears

²⁰Compare the standard Bayesian claim that apparently unconditional probabilities are always implicitly conditioned on an agent’s evidence (e.g., Jaynes 2003).

²¹The intuitions described in the main text discussion rely on the assumption that ϕ is defined everywhere in c . Some subtleties arise because c_ϕ includes worlds where ϕ is true or undefined. With left-nested conditionals *if* (ψ *if* ϕ) χ , this means that worlds where the embedded antecedent ϕ is false or undefined will also remain. It is difficult to judge whether this is problematic given the marginal status of left-nested conditionals in general. See Égré et al. 2025b for relevant discussion of the probabilities of nested conditionals.

to be correct. In disjunctions, for example, expressions of probability in a right disjunct are indexed to a probability measure that is effectively conditioned on the falsity of the left disjunct. Consider a fair six-sided die that has been tossed, but we do not know the result:

- (48) Either the die came up odd, or there's a $2/3$ probability that it was greater than 3.

The probability measure relevant for evaluating the right disjunct is the *conditional* probability that the die was greater than three, given the *falsity* of the left disjunct. This behavior is exactly what we would expect if Stalnaker's Thesis holds because of implicit indexing of the probability measure to a local context in which the falsity of the first disjunct is taken for granted. Note that the same holds whether we consider a probability expressions in an unembedded or an embedded environment. The resulting interpretation is exactly as if there were an implicit conditional in the right disjunct.

- (49) Either the die came up odd, or—if it came up even—there's a $2/3$ probability that it was greater than 3.

4.3 Update: Indicative conditional = Material conditional

Here the temporary-context theory again inherits a positive feature of the trivalent semantics. An assertion of χ in c is a proposal to remove worlds in c that make χ false. So, an assertion of *If* ϕ, ψ is a proposal to remove from c worlds where ϕ is true and ψ is false. For bare conditionals, where ψ does not contain any material affected the context shift due to *If* ϕ , this update is the same as update by $\neg(\phi \wedge \neg\psi)$, which is equivalent to the material conditional $\phi \supset \psi$. This derivation accounts for Khoo's (2022) first Bounding Puzzle.

4.4 No silent operators

No silent operators are needed in the temporary-context theory. Instead, the interpretation of every sentence in the language is a function from possible worlds to $\{0, 1, \#\}$, and the pragmatic theory is equipped to handle each of these values.

4.5 Domain restriction & presupposition

In this section we show that the temporary-context theory can derive the use of conditionals to restrict modal domains, on several plausible theories of modal semantics. Since the main focus of this paper is not on modal semantics per se, we will not attempt to canvas the full range of available theories of modality or develop one theory in detail. Instead, the theories discussed below are chosen for familiarity and to illustrate our main point: as long as modal domains are appropriately tied to local contexts, conditional restriction of modal domains comes for free in the temporary-context theory.

Note, however, that the theory does not immediately account for the use of conditionals to restrict adverbs of quantification. I return to this point briefly in §6.2 below.

4.5.1 A trivalent domain semantics

A straightforward approach to conditional restriction of modal domains is to adapt the bivalent domain semantics discussed in §2.3 to a trivalent setting, treating the strong modal as a generalized version of *and* as defined in (36). Here *BEST* is a function taking a set of worlds to the best worlds in the set, according to some contextually relevant norms.

- (50) $\mathcal{I}(\ulcorner \Box_e(\phi) \urcorner, c, w) =$
- a. 0 if there is a $w' \in c$ such that $\mathcal{I}(\ulcorner \phi \urcorner, c, w') = 0$;
 - b. $\#$ if, for all $w' \in c$, $\mathcal{I}(\ulcorner \phi \urcorner, c, w') = \#$;
 - c. otherwise, $= 1$.

- (51) $\mathcal{I}(\ulcorner \Box_d(\phi) \urcorner, c, w) =$
- a. 0 if there is a $w' \in \text{BEST}(c)$ such that $\mathcal{I}(\ulcorner \phi \urcorner, c, w') = 0$;
 - b. $\#$ if, for all $w' \in \text{BEST}(c)$, $\mathcal{I}(\ulcorner \phi \urcorner, c, w') = \#$;
 - c. otherwise, $= 1$.

The weak modals $\Diamond_e$ and $\Diamond_d$ are defined as the duals of their respective strong items: $\Diamond_x\phi := \neg\Box_x\neg\phi$.

With these definitions in hand, the temporary-context theory captures domain restriction of modals in exactly the same way as Restriction of Local Contexts approach discussed in section 2.3 (Klinedinst & Rothschild, 2012; Yalcin, 2007). On this approach, modals are sensitive to conditional antecedents not because they are lexically specified as such, but because their domain is provided by the local context of evaluation. As a result, the domain of a modal occurring in the consequent of a conditional *If* ϕ , *MODAL* ψ , evaluated in context c , will be the local context of the consequent, c_ϕ . This explains why conditional antecedents restrict modal operators. It also explains at the same time why presuppositions in the consequent of a conditional can be filtered by the antecedent, yielding conditional presuppositions like (12).

For instance, consider a conditional restricting an epistemic modal as in *If Sal isn't at home, she must be at the movies*, with *must* translated as $\Box_e$ as defined in (50).

- (52) $\mathcal{I}(\ulcorner \neg\text{home} \Rightarrow \Box_e\text{movies} \urcorner, c, w) = \mathcal{I}(\ulcorner \Box_e\text{movies} \urcorner, c_{\neg\text{home}}, w)$
- a. = 0 if there is any $w' \in c_{\neg\text{home}}$ such that $\llbracket \text{movies} \rrbracket^{c_{\neg\text{home}}, w'} = 0$;
 - b. = # if, for all $w' \in c_{\neg\text{home}}$, $\llbracket \text{movies} \rrbracket^{c_{\neg\text{home}}, w'} = \#$;
 - c. = 1 otherwise.

The modalized conditional is false if there are any worlds in the restricted context $c_{\neg\text{home}}$ in which Sal is not at the movies; true if Sal is at the movies in all worlds in the restricted context $c_{\neg\text{home}}$; and undefined/ignorable in one special case. (52) has the value # (at any world) whenever **movies** is # at every world w' in the restricted context $c_{\neg\text{home}}$. Since **movies** is atomic, this can occur only when every such w' fails to be in $c_{\neg\text{home}}$ —in other words, when restricting c to worlds where Sal is not at home also eliminates all worlds in which she is at the movies. In effect, the conditional is predicted to be # at every world—and so unassertible—if the context does not allow that Sal might be not-at-home and also at the movies.

The predicted behavior of deontic modals is similar, except that the domain is further restricted to the best worlds in the restricted context. The corresponding assertibility condition in this case is that the context may not exclude the possibility that the antecedent and consequent both hold.

The story about disjunction is essentially the same, and is familiar from §2.3. The local context of a right disjunct, evaluated in c , is the global context restricted by the negation of the left disjunct. As a result, presuppositions are filtered by the negation of the left disjunct, and modals in the right disjunct have a domain that includes the negation of the left disjunct. For example, *Either Sal is at home, or she must be at the movies* will be interpreted as

- (53) $\mathcal{I}(\ulcorner \text{home} \vee \Box_e\text{movies} \urcorner, c, w) = 1$ iff either $\mathcal{I}(\ulcorner \text{home} \urcorner, c, w) = 1$ or $\mathcal{I}(\ulcorner \Box_e\text{movies} \urcorner, c_{\neg\text{home}}, w)$.

The rest of the derivation proceeds exactly as in (52). The only semantic difference between the conditional and disjunctive variants is that the disjunction is true in worlds where Sal is at home, regardless of what is going on in the rest of the context.

Other, less well-known ways of modifying local contexts lend further support to the idea that modal domain restriction is tied to context-shifting operators. For example, *supposing*- and *assuming*-clauses presumably operate in the same way as *if*-clauses, updating the context parameter to incorporate a temporary assumption. The discourse-anaphoric particle *otherwise* has a similar effect, temporarily adding an assumption to the effect that some previously mentioned proposition is false (Webber, Stone, Joshi, & Knott, 2003). As expected, all of these expression-types restrict the domains of modals in the same way as *if*-clauses.

- (54) Domain restriction with *assuming* and *otherwise*
- a. Assuming Sal isn't at home, she must be at the movies.
 - b. Sal is probably at home. Otherwise, she must be at the movies. [= "If she is not at home, ..."]

They also filter presuppositions in an identical fashion to *if*-clauses.

- (55) Presupposition filtering with *assuming* and *otherwise*
- a. Assuming Sal is a diver, she forgot her scuba gear.
 - b. I guess Sal isn't a diver. Otherwise, she forgot her scuba gear. [= "If she is a diver, ..."]

These more general phenomena confirm that the temporary-context theory is correct to decouple modal domain restriction from *if*-clauses per se, and to give this job instead to a parameter representing contextual information that also controls presupposition satisfaction, and can be modified by various linguistic devices.

4.5.2 A trivalent relational semantics

The domain semantics treats modal claims as depending on the context of evaluation, independently of the world of evaluation. This may or may not be a desirable feature. For deontic modals, the prediction of sensitivity to contextually available information is highly plausible in some cases, but problematic in others (Cariani, Kaufmann, & Kaufmann, 2013; Chrisman, 2015). In particular, the domain semantics in (51) would seem to rule out the truth of strong deontic claims whose prejacentes are known to be false, and therefore are excluded by the context.

- (56) a. There ought to be no war.
b. You ought to leave, but you we both know you won't.

In addition, tying deontic claims to a fixed domain makes it difficult to account for normative uncertainty, in which worlds that are contextually open disagree about what the relevant norms are.

- (57) a. Either you ought to wash the dishes, or you ought to clean your room, but I don't recall which. (I'll ask our parents what they said.)
b. If your mother said to wash the dishes, you have to wash the dishes. (Did she?)
c. If Kant is right, we must never lie.

One response to these and similar issues is to employ a variant of the relational semantics in which modal domains are sensitive to both the world of evaluation and the local context. (For these purposes, Kratzer's theory of modality can be considered as a variant of the relational semantics.)

We can adapt the classical relational semantics to a trivalent setting by adding two parameters R_e and R_d for the epistemic and deontic accessibility relations, and defining the modal operators as generalized conjunction/disjunction as above.

- (58) $\mathcal{I}(\ulcorner \Box_{e/d} \phi \urcorner, c, w, R_e, R_d)$
a. = 0 if there is a $w' \in R_{e/d}(w)$ such that $\mathcal{I}(\ulcorner \phi \urcorner, c, w') = 0$;
b. = # if, for all $w' \in R_{e/d}(w)$, $\mathcal{I}(\ulcorner \phi \urcorner, c, w') = \#$;
c. otherwise, = 1.

In contrast to the domain semantics given above, the interpretation of modals in the relational semantics depends on the world of evaluation. Given the broader assumptions of the temporary-context theory, this implies that a modalized conditional has the value # in all worlds where its antecedent is false. A modalized conditional is thus interpreted as follows:

- (59) $\mathcal{I}(\ulcorner \phi \Rightarrow \Box_e \psi \urcorner, c, w, R_e, R_d) = \mathcal{I}(\ulcorner \Box_e \psi \urcorner, c_\phi, w, R_e, R_d)$
a. = # if either
i. $w \notin c_\phi$, or
ii. for all $w' \in R_e(w)$, $\mathcal{I}(\ulcorner \psi \urcorner, c_\phi, w', R_e, R_d) = \#$;
b. = 0 if there is a $w' \in R_e(w)$ such that $\mathcal{I}(\ulcorner \psi \urcorner, c_\phi, w', R_e, R_d) = 0$;
c. otherwise, = 1.

As usual, we need to impose further constraints on R_e and R_d in order to get sensible interpretations. For instance, it is usual to assume that epistemic accessibility is an equivalence relation (reflexive, symmetric, and transitive). Since the context is meant to represent live epistemic possibilities, it is natural to assume further that at context c , the accessible worlds from each world in c are simply c itself: $R_e(w) = c$ for each $w \in c$.²² On this assumption, consider again

²²Note that we are assuming only that the global context of the sentence constrains R_e . This is weaker than Mandelkern's (2019) assumption that the epistemic accessibility relation changes in the course of interpretation along with changes in the local context. It would be straightforward to add this as an additional constraint in order to capture data around epistemic contradictions and related phenomena that Mandelkern discusses. Detailed examination of the predictions of this move and its interaction with the temporary-context theory will have to await another occasion, though.

(60) If Sal isn't at home, she must be at the movies.

Relative to context c , this is interpreted as follows. First, we divide c into worlds where the antecedent ϕ is false, which receive the value $\#$ since the consequent is evaluated in c_ϕ . (For the purpose of assertibility, these worlds are then ignored. No other worlds in c receive $\#$ by clause (59a-ii), because the consequent is atomic and so bivalent.) Second, we partition the remaining worlds—where the antecedent ϕ is true—into two sets. At worlds w where there is a $w' \in R_e(w)$ at which the modal prejacent *Sal is at the movies* is false relative to c_ϕ , (60) receives the value 0. In the remainder, where the prejacent is either true or $\#$ at every accessible world, (60) receives the value 1.

Perhaps surprisingly, this is enough to account for conditional restriction of epistemic modals. Since all worlds in the context c can see the same accessible worlds—namely, c itself—the worlds where the antecedent is true will all be lumped together: if any of them make *Sal is at the movies* false, then they will all receive the value 0. Otherwise, all receive the value 1. In the latter case, each true-antecedent world is associated with a modal domain that also includes the false-antecedent worlds in c . But in these worlds, the value of the prejacent is $\#$, since they are excluded by the context c_ϕ . This is enough to render (60) true by the definition in (58). As a result the interpretation of the modal in (60) depends only on whether the prejacent is true throughout the true-antecedent worlds: true if so, false otherwise.

The effect is similar to Huitink's (2008) theory that we discussed above: the modal is true just in case the prejacent is either true or $\#$ at every world in its domain. However, the temporary-context theory is at an advantage because domain restriction is not due to a lexical stipulation about the modal's handling of $\#$ values. Instead, it results from the interaction of two independently motivated features of the theory: (i) conjunction treats $\#$ values as ignorable, and (ii) strong modals pick out generalized conjunction as defined above (§3.1). Together, these conditions imply that conditional antecedents restrict the domains of epistemic modals in their consequents. There is no need to stipulate this sensitivity in the lexical entries of modals.

Deontic modals are somewhat more complex, because they vary in their sensitivity to contextual information. Following Cariani et al. (2013), we can distinguish “objective” and “deliberative” interpretations of deontic modals. The former depend only on what is ideal according to relevant norms, while the latter are also sensitive to contextually relevant information. The examples in (56) and (57) above are instances of objective interpretations. (56) illustrates insensitivity of the domains of deontic modals to contextual information: e.g., the fact that the world is imperfect does not render “There ought to be no war” trivially false. The examples in (57) are cases of deontic uncertainty. In brief, worlds in the context disagree about what is optimal: at some worlds it's optimal to wash the dishes (because that's what your mother said to do), and at other worlds it's optimal to clean your room (because that's what your mother said to do). We just don't know which kind of world we are in.

The relational semantics handles the objective reading of deontic claims straightforwardly. Each world in the context is associated with a possibly different set of accessible worlds, and there is no analogue of the epistemic assumption that all and only worlds in the context are accessible. Consider a deontic conditional with an intuitively objective interpretation such as (61). Here, the antecedent of the conditional does *not* restrict the domain of the modal: instead, it tells us what the facts of the world would have to be like for the consequent to be true.

(61) If your mother said to wash the dishes, you have to wash the dishes.

For familiar reasons, this sentence is $\#$ at any world in which your mother did not say to wash the dishes. (As usual, it may still be *assertible* at such worlds, as long as the relevant context allows that she *might* have said this.) At any world w in which she did say this, the sentence is true just in case, at every deontically accessible world $w' \in R_d(w)$ in which the antecedent is true, the prejacent is true as well. However, (61) does not imply anything about what is optimal in worlds in which the antecedent is false. The Kant example (57c) yields to a similar analysis.

On the deliberative reading of deontic modals, by contrast, the domain of quantification is sensitive to contextual information in a similar way to epistemic modals. This is also the interpretation on which conditional antecedents restrict deontic modals, as discussed by Kratzer (1991b) and in much following work. For example:

(62) If you offended Bill, you should apologize.

The intuitive interpretation of (62) implies nothing about what is optimal, relative to any possible world. Indeed, it is plausible that you would apologize in *none* of the optimal worlds, because in these worlds you

have done nothing in need of apology. On this assumption, *You should apologize* is false at every world on the objective interpretation (unless it is # everywhere and therefore trivial). Yet (62) is sound advice in many contexts. To get the right interpretation of (62), what we need is an information-dependent interpretation of *should* which can be affected by the conditional antecedent.

There are several ways this could be done. One way, inspired by Kratzer (1991b), is to add an “ordering source” which takes each world w to a pre-order $\succsim_w$ interpreted as “deontically as good as”. This pre-order is restricted to worlds in the local context. On this interpretation, deontic modals have domains that are determined not by what is optimal in general, but by what is optimal among the relevant possibilities left open by the local context. Since we are assuming an objective/information-sensitive ambiguity in deontic modals, I will notate the information-sensitive reading with a third modal $\Box_i$, and I will simplify relative to Kratzer’s theory by treating $\succsim_w$ as a further parameter of evaluation.

Define $w' \succ_w w''$ as $w' \succsim_w w'' \wedge \neg(w'' \succsim_w w')$, and following Kratzer let $BEST(w, c) = \{w' \in c \mid \neg \exists w'' \in c : w'' \succ_w w'\}$. Intuitively, the best worlds relative to (c, w) are the worlds in c for which no world in c is deontically better by the lights of w . We then define the information-sensitive strong deontic modal as:

- (63) $\mathcal{I}(\ulcorner \Box_i \phi \urcorner, c, w, R_e, R_d, \succsim_w)$
- a. = 0 if there is a $w' \in BEST(w, c)$ such that $\mathcal{I}(\ulcorner \phi \urcorner, c, w') = 0$;
 - b. = # if, for all $w' \in BEST(w, c)$, $\mathcal{I}(\ulcorner \phi \urcorner, c, w') = \#$;
 - c. otherwise, = 1.

The information-sensitive deontic modal, unlike its objective variant, depends on contextually available information. On this definition, *You should apologize* can be true at a world and context in which it is known that you offended someone, even if the best possible worlds are ones in which you never offended anyone (and so have no need to apologize).

We also expect that the domains of embedded information-sensitive modals will depend on the local context in which they occur. In a conditional *If ϕ , $\Box_i \psi$* , uttered in c , the domain of $\Box_i$ will be restricted to worlds in which the antecedent is true, because $\Box_i$ has a domain restricted to its local context c_ϕ . This accounts for domain restriction of the deontic modal in (62), which is true if and only if the best worlds *among those in which you offended Bill* are worlds in which you apologize. Analogous considerations account for examples of deontic domain restriction in disjunctions reviewed in section 2.2 and 2.3 above.

4.5.3 Upshot on domain restriction

After this fairly detailed examination of two ways to capture domain restriction of modals, let’s take stock. The purpose of this section was not to advocate for either of the theories described here. Indeed, both have limitations as a general theory of modal semantics in my opinion. Rather, the point was to illustrate that there are numerous plausible theories of modality on which domain restriction of modals by conditionals and disjunctions emerges as an automatic byproduct of context-shifting, with no need to stipulate these facts as a special feature of conditionals. The reader is invited to explore the predictions of a favored theory of modality when plugged in here: say, one built around probability and expected value, or around reasons and evidentiary relations.

5 The explanatory challenge

The temporary-context theory is designed to capture the advantageous features of various previous accounts—the restrictor theory, the Restriction of Local Contexts theory, and the trivalent semantics—while avoiding their drawbacks. In its logical behavior, the temporary-context theory is fundamentally a variant of the trivalent semantics, but it has certain advantages over previous accounts in this vein. It captures the core phenomena motivating the trivalent theory while also explaining domain restriction, presupposition projection, and the parallels between them. In addition, the temporary-context theory responds to the explanatory challenges discussed for trivalent theories in §2.4.

Why are conditionals associated with a third value? In the temporary-context theory, the answer is not that conditionals are special. Rather, the third value is assigned to p at all and only (w, c) pairs where $w \notin c$ —in other words, when the context ensures that the world in question is “ignorable”. *If*-clauses are one way of modifying a context, and one that can reliably cause worlds in the global context to take on the # value. The interpretation of *or* also involves context manipulation, but the fact that

the truth of the left disjunct ϕ is sufficient for the truth of $\phi \vee \psi$ hides the presence of the $\#$ value in the evaluation of the right disjunct ψ .²³ We can, however, detect the shift by its effects on presuppositions and modal domains.

Other devices introducing suppositions also induce $\#$ -values. Above we discussed *assuming*, *supposing*, and *otherwise*, but there are many more. Finally, there are context-shifting operations that are not intersective with the global context, such as the evaluation of subjunctive conditionals. While the semantic and pragmatic issues around these constructions are complex enough to deserve a separate paper, their behavior with respect to modal domain restriction and presupposition is consistent with the claim that these phenomena depend on operations that shift local contexts in various ways (see brief discussion in §6.1 below).

6 Further properties and challenges

The temporary-context theory has a number of further attractions that we have not discussed here, primarily for reasons of space. In particular, it is readily integrated with previous accounts that also draw inspiration from Kaufmann (2000), handling a wide range of phenomena including modal subordination, conditional questions, and more (Biezma, 2025; Bledin & Rawlins, 2019; Isaacs & Rawlins, 2008). It is also promising with respect to the treatment of other non-declarative conditionals, interactions between *if*-clauses and nominal quantifiers, and the semantics of conditionals embedded under attitude verbs and in other contexts. Needless to say, many further phenomena remain to be explored.

6.1 Subjunctive conditionals

Subjunctive conditionals are another major topic that we have not discussed in detail. The most natural approach, from the perspective of the temporary-context theory, is to treat the morphological marking of a subjunctive conditional as introducing a selection function f that takes a sentence A and a world of evaluation w to a new world $f(A, w)$ in which A is true, while minimizing other disruptions. This is simply Stalnaker’s (1968) theory, but limited to subjunctives; it is also close in spirit to the treatment of subjunctive conditionals based on causal models in Pearl 2000. Such a theory would also need to explain how presupposition filtering and modal domains are sensitive to the derived context of a subjunctive conditional. For instance, (64a) presupposes (64b). This suggests that the local context of the consequent is one in which the relevant selection function has mapped every world in the input context to a world in which the speaker is a diver. The presuppositional constraint that the sentence places on the global context is that, in each of the resulting worlds, the speaker also has scuba gear.

- (64) a. If I were a diver, I would not have forgotten my scuba gear.
 b. $\rightsquigarrow$ If I were a diver, I would have scuba gear.

Similarly, the quantificational domain of a modal in the consequent of a subjunctive conditional appears to include subjunctive counterparts of each world in the input context. For example, consider a context in which Bill has just awoken from a long sleep and does not know what day it is. So, the context includes worlds where it is Monday, worlds where it is Tuesday, and so on, each day with equal probability. Feeling unkempt, he considers trying to get a haircut, but reflects that his favorite barbershop is closed Sundays and Mondays, and decides that it’s too much trouble. After all:

- (65) If I were to go to the barbershop, there’s a 2/7 chance that I would be disappointed.

The truth of (65) follows in this context if each world in the input context is mapped, in a probability-preserving manner, to a subjunctive counterpart in which the day remains the same and Bill goes to the barbershop. In 5/7 of these he gets a haircut, but in 2/7 he goes home without one.

A temporary-context theory of subjunctive conditionals may be able to account for these patterns by defining the derived context of a subjunctive conditional with antecedent ϕ as doing two things: shifting the world of evaluation, and mapping the function $\lambda w.f(\phi, w)$ across the input context.

²³That is, if $w \in c$ and ϕ and ψ are both true at w , then $\mathcal{I}(\ulcorner \phi \vee \psi \urcorner, c, w)$ is true because $\mathcal{I}(\ulcorner \phi \urcorner, c, w) = 1$, even though the right disjunct is evaluated as $\mathcal{I}(\ulcorner \psi \urcorner, c_{-\phi}, w) = \#$ because $w \notin c_{-\phi}$. Once we know that the left disjunct is true, the right disjunct can be ignored because the disjunction is true no matter what. This is closely related to Schlenker’s (2009) considerations around the “smallest domain that one may restrict attention to when assessing” an expression (see further discussion in §6.3 below). It also has a suggestive similarity to lazy functional programming languages such as Haskell, where a right disjunct is not evaluated if the left disjunct is true, but the converse does not hold (e.g., Hughes 1989; Hutton 2016: §15).

$$(66) \quad \mathcal{I}(\ulcorner \text{if } \phi, \text{ would } \psi \urcorner, c, w) = \mathcal{I}(\ulcorner \psi \urcorner, c', f(\phi, w)), \text{ where } c' = \{f(\phi, w') \mid w' \in c\}.$$

Notice that the interpretation of the *if*-clause in (66) will never lead a subjunctive conditional to be undefined. This is because the only source of undefinedness in the present theory is for the world of evaluation to fail to be in the local context. However, a subjunctive conditional, interpreted as in (66), will have a derived evaluation world $f(\phi, w)$ that is also an element of the derived context $\{f(\phi, w') \mid w' \in c\}$ as long as $w \in c$. This is in contrast to indicative conditionals, where the evaluation world can fail to be in the derived context because the latter may be restricted to exclude it.

The dual role of the selection function in this approach would explain both how the antecedent influences truth-value intuitions and why the antecedent is true throughout the local context of a subjunctive conditional as well. In addition, when combined with the approach to context-sensitive probability calculation advocated above, it may provide a compositional approach to the problem of computing subjunctive probabilities. Naturally, much more work would be required for this sketch to be turned into a complete theory of subjunctive conditionals.

6.2 Adverbs of quantification

The temporary-context theory does not immediately extend to the use of conditionals to restrict adverbs of quantification, as in (67).

(67) If Caesar awoke early, he usually went for a walk.

Usually is not a modal: it quantifies over a class of events, situations, or some such that exist in the actual world. The relevant notion of restriction in (67) does not involve restricting a set of alternative worlds considered as live possibilities, but restricting a set of real-world events or situations to those with the property picked out by the antecedent. As a result, the temporary-context theory would need substantial further development in order to account for (67).

On face, this seems like a major lacuna. After all, the first paper pointing to domain restriction by conditionals, [D. Lewis 1975](#), only discussed adverbs of quantification. It was [Kratzer \(1991b\)](#) who added modals into the mix and suggested that there should be a unified theory in which conditional antecedents function as generalized devices of domain restriction. Most of the subsequent literature has focused on modals, but it is often assumed that the two phenomena have a unified explanation.

Adverbs of quantification are usually thought to quantify over more fine-grained objects than possible worlds—world/assignment function pairs, perhaps, or situations construed as parts of worlds. It is possible that the temporary-context theory could be extended to the adverbial case if the representation of context that it employs were enriched to make use of these objects. Needless to say, this would be a substantial technical and empirical project that is best reserved for another occasion.

Alternately, it may turn out that conditional restriction of adverbs of quantification relies on a different mechanism from restriction of modals. Adverbs of quantification are sensitive to a much wider range of linguistic and pragmatic factors than modal domains. This suggests that their domains may be determined differently. For instance, *when*-clauses can restrict adverbs, but not modals.

- (68) a. When Caesar awoke early, he usually went for a walk. ($\neq$ Caesar usually went for a walk.)
b. When Sal awoke early, she must have gone for a walk. ($\models$ Sal must have gone for a walk.)

(68b) is sensible, but it differs from the restrictive *when*-clause in (68a) in several respects. For instance, (68a) does not entail the truth of its main clause, but (68b) does. This suggests that the *when*-clause in (68b) has the ordinary temporal meaning that you see in (69): it picks out the time at which the modal prejacent is said to hold.

(69) When Sal awoke early this morning, she went for a walk. ($\models$ Sal went for a walk.)

Adverbs of quantification are also sensitive to other linguistic factors that do not affect modals, or affect them differently. For instance, [Beaver and Clark \(2008\)](#) discuss the focus-sensitivity of quantificational adverbs in great detail, arguing that it occurs because focus placement reflects the current Question Under Discussion (QUD: [Ginzburg 1995](#); [Roberts 2012](#); [van Kuppevelt 1995](#)). The QUD, in turn, helps in determining which situations/cases are pragmatically relevant to the adverb's quantificational force.

(70) Adverbs of quantification:

- a. Mary usually invites ANDY to parties.
 ≈ “If Mary invites someone to a party, it’s usually Andy.”
- b. Mary usually invites Andy to PARTIES.
 ≈ “If Mary invites Andy somewhere, it’s usually to a party.”

Modals also show limited focus-sensitivity, but it does not have analogous effects on the interpretation of the modal.²⁴

(71) Modals:

- a. Mary must have invited ANDY to the party.
 ≠ “If Mary invited someone to the party, it must have been Andy.”
- b. Mary must have invited Andy to the PARTY.
 ≠ “If Mary invited Andy somewhere, it must have been to a party.”

Beaver and Clark (2008) also discuss the lesser-known phenomenon of *Association with Presupposition*, where an adverb’s domain is restricted to situations that satisfy presuppositions within its scope. For example, consider cases such as (72) (originally pointed out by Schubert and Pelletier 1987; see also Calcagno 1996; Cohen 1999).

- (72) a. Mary always beats Sandy at ping-pong.
 ≈ “If Mary and Sandy play ping-pong, Mary always wins.”
- b. This cat usually lands on its feet.
 ≈ “If this cat falls from a height, it usually lands on its feet.”

Here again, modals are different: their domains are not affected by presupposition in an analogous way.

- (73) a. Mary must have beaten Sandy at ping-pong.
 ≠ “If Mary and Sandy played ping-pong, Mary must have won.”
- b. This cat must have landed on its feet.
 ≠ “If this cat fell from a height, it must have landed on its feet.”

Cohen (1999) shows that the effects of association with presupposition on adverbial domains can override the effects of prosodic focus. Beaver and Clark (2008: §8) discuss this and related observations in detail, concluding that these points support a theory in which the domains of adverbs of quantification are determined by “a pragmatic and inferential process”. This process involves integrating possibly conflicting cues from surrounding linguistic, discursive, and pragmatic context in order to infer a domain that renders the resulting interpretation coherent and sensible.

Conditional antecedents may turn out to be just another cue that plays into the complex pragmatic process of determining the domains of adverbs. In contrast, the temporary-context theory concurs with the mainstream position that conditionals restrict modal domains as part of the compositional semantics. This contrast would help to explain why adverbial domains are sensitive to numerous factors which do not have analogous influences on modals. A compositional story about adverbs of quantification might be feasible within the temporary-context theory, as noted above. However, the fact that it does not immediately provide one could turn out to be a blessing in disguise.

6.3 Schlenker’s explanatory challenge for a theory of local contexts

Throughout this paper, the temporary-context theory has been advertised as a way to account for the basic phenomena around conditionals in parsimonious and explanatory way, unifying theoretical considerations from various domains within a single, relatively simple formal theory. There is, however, an apparent conflict between the explanatory goals of the temporary-context theory and one of its acknowledged inspirations: the theory of local contexts in Schlenker 2009. This section explains the problem and shows that it is merely apparent. Even though the temporary-context theory relies on the

²⁴In terms of Beaver and Clark’s (2008) three-way taxonomy of types of focus-sensitivity, the data in (70)-(71) suggest that modals primarily display quasi-association with focus, which involves the placement of focus indicating what the speaker takes the current QUD to be, with concomitant pragmatic effects. As they argue in detail, quantificational adverbs show free association with focus, where the QUD also helps the listener to infer the intended domain. Beaver and Clark (2008: 61-2) discuss the possibility that *must* also shows free association. However, their key example (*Dogs must be carried*) is probably best analyzed as a generic scoping over entire clause, and generics clearly show quasi-association. The examples in (71) argue directly against a free association treatment of focus-sensitivity for *must*.

possibility that an expression can change the context as part of its lexical semantics, this is compatible with a non-stipulative theory of local contexts in general.

Schlenker argues that dynamic semantic theories following Heim 1983, which stipulate the dynamic effects of connectives as part of their lexical semantics, are non-explanatory. For example, in dynamic semantic theories there is an asymmetry between the left and right conjunct in a conjunction ϕ and ψ , as uttered in context c . The local context of ϕ is the global context c , while the local context of ψ is c_ϕ , the global context updated with the truth of the left conjunct. This is meant to account for the fact that presupposition filtering in conjunctions is asymmetric: a left conjunct can license a presupposition in the right conjunct, but not vice versa.

- (74) a. Mary is a diver, and she forgot her wetsuit.
b. Mary forgot her wetsuit, and she is a diver.

The key observation is that (74a) is felicitous in a context in which it is not known in advance that Mary has a wetsuit, as long as it is known that divers generally have wetsuits. By contrast, (74b) requires a context in which it is known that Mary has a wetsuit.

As Soames (1982) points out and Schlenker (2009, 2010) discusses in detail, this account fails to explain *why* filtering is asymmetric with *and*. In dynamic semantic theories, it is straightforward to encode the relevant pattern as described above. But it is just as easy to state a variant *and*^{*} which reverses the pattern—one in which the local context for the right conjunct ψ is the global context c , and the local context of the left conjunct is the global context updated with the right. This would predict the opposite pattern of judgments to (74). What, then, explains why *and* is attested and *and*^{*} is not?

The criticism applies equally to the static variants in Klinedinst and Rothschild 2012; Yalcin 2007 that we used as a starting point for the temporary-context theory. In these theories, and in the temporary-context theory as stated above, the local context shifts associated with the connectives and *if* are stipulated in the lexicon, rather than being derived from general principles.

Schlenker (2009) argues that a principled theory of local contexts should be able to predict, based on general theoretical considerations and the bivalent lexical entries of the connectives, how they shift local contexts. In brief, he suggests that the local context of an expression, evaluated at c , is the smallest portion of c to which an interpreter can restrict attention without risk of making an error in assessing the sentence. Combined with the assumption that semantic evaluation proceeds left-to-right, this approach predicts that the local context of ϕ in ϕ and ψ is c , while the local context of ψ is c_ϕ . The intuition is that, once we have evaluated $\lceil \phi \text{ and } \rceil$, we already know that the sentence will be false at $\neg\phi$ -worlds. As a result, it does no harm to restrict attention to the ϕ -worlds in evaluating the rest of the sentence, because we already know how to classify the others. Schlenker (2009) shows that similar reasoning accounts for disjunction: in ϕ or ψ , the fact that the local context of ψ is $c_{\neg\phi}$.²⁵

Schlenker’s theory also accounts for the fact that the local context of a conditional’s consequent is the global context updated with the antecedent. However, this result follows only because Schlenker analyzes *if* as the material conditional. Since the material conditional can be evaluated as true in false-antecedent worlds without considering the consequent, a lazy evaluator can restrict attention to true-antecedent worlds while evaluating the consequent. Thus, Schlenker’s algorithm correctly predicts that the local context of ψ in $\phi \supset \psi$ is c_ϕ .

However, this derivation comes at a serious cost: it requires assuming a semantic theory of *if* which is known to be inadequate in numerous respects (cf. discussion in §2.1). A natural question here is what would happen if we were to adopt a standard bivalent, modal theory of conditionals. In fact, this move would disrupt the reasoning behind Schlenker’s derivation. In modal theories, the conditional can be either true or false in worlds in which the antecedent is false. Therefore, a lazy evaluator cannot safely ignore worlds in which the antecedent is false while evaluating the consequent. As a result, a bivalent modal theory, combined with Schlenker’s theory of conditionals, incorrectly predicts that the local context of a conditional consequent should be the same as the global context: no presupposition filtering in conditionals. If we take Schlenker’s explanatory challenge seriously, then, this should count as a further objection to modal theories of conditionals.

²⁵Here again, the reasoning bears a close conceptual affinity to lazy evaluation strategies employed in functional programming languages. For instance, the Haskell interpreter returns *False* for a conjunction with a false left conjunct, without evaluating the right conjunct—and so can return a result even if evaluating the right conjunct would require constructing an infinite list. Similarly, it returns *True* for a disjunction with a true left disjunct, without evaluating the right disjunct; and the consequent of a conditional is not evaluated if the antecedent is false. As noted above (fn. 23), this style of reasoning also motivates the way that the “ignorable” third value is handled by connectives and modals in the temporary-context theory.

At this point, there are two different ways to meet Schlenker’s challenge. The first is to return to standard trivalent theories (§2.4), in which false-antecedent conditionals have the # value by lexical stipulation. On this approach, we can explain local context shifting within Schlenker’s general approach straightforwardly: worlds where a conditional antecedent is false can be classified immediately as # without evaluating the consequent. As a result, there is no semantic risk involved in restricting attention to the true-antecedent worlds when evaluating the consequent.²⁶ This approach seems promising, but it is subject to the same objections that we raised against trivalent theories above with respect to domain restriction and presupposition projection. Moreover, if this were the only solution, then Schlenker’s explanatory challenge would be in competition with the explanatory challenge for trivalent theories introduced in this paper (§2.4).

We can address both explanatory challenges, however, if we adopt the variant of Schlenker’s theory proposed by Barker (2022). Barker shows that a continuation-based semantics can capture Schlenker-style reasoning about local contexts as an automatic byproduct of incremental interpretation, starting with bivalent definitions of the connectives. However, Barker’s theory has a crucial degree of freedom that Schlenker’s lacks: in the continuation-based theory, it is possible for expressions to be lexically specified to override the normal rules by which local contexts evolve. Schlenker’s key results remain as defaults when expressions do not lexically specify any changes to the context. Barker (2022: §7) argues that this additional degree of freedom is needed to account for cases in which the default projection algorithm would otherwise yield the wrong result.

Barker’s framework leaves technical space for lexical adjustment of local contexts, and he argues that languages do make use of this feature. From this perspective, there is no issue in adopting the same approach to *if* and other suppositional language. We simply provide *if* with a lexical entry that takes an input context c and two propositional arguments ϕ and ψ , and returns the result of evaluating ψ at context c_ϕ . From this perspective, there is no conflict between Schlenker’s explanatory challenge and our goal of explaining why a trivalent theory of meaning has the structure that it does. Some expressions’ meanings are defined in terms of context changes; for the rest, context shifts happen as an automatic byproduct of lazy semantic evaluation.

7 Conclusion

In light of the failings of the material conditional, most theorists have concluded that conditionals have a modal semantics. Much of the debate has been around whether they express strict conditionals, variably strict, or if they are domain restrictors whose modal force derives from a separate operator. However, modal theories face a variety of empirical and conceptual problems. For instance, most modal theories predict excessively weak interpretations of negated conditionals and make plainly incorrect predictions about the probabilities of conditionals. In addition, many modal theories fail to provide a unified account of domain restriction by conditionals and disjunction.

Historically, the shift to a modal semantics for conditionals was motivated by a false dichotomy. The failure of the only plausible *bivalent* truth-functional theory is not sufficient reason to conclude that no truth-functional theory of *if* is possible. Trivalent theories, where conditionals with false antecedents receive a third semantic value, turn out to do extremely well on numerous points: they interact well with negation, predict correct probabilities for conditionals, and can account for the domain-restricting effects of conditionals. However, it is less apparent how to explain the domain-restricting effects of disjunction in this framework, and more generally parallels between presupposition projection and domain restriction. In addition, the trivalent semantics faces several explanatory challenges: why would conditionals be special in making use of a third value? And why do probability and modal judgments ignore this value, rather than grouping it with True or False?

The temporary-context theory responds to this explanatory challenge while also maintaining the desirable features of modal theories. The key assumptions are (1) that every sentence is evaluated against a local context, as in popular theories of presupposition; (2) that semantic and pragmatic interpretation does not classify worlds outside the local context of evaluation as true or false, treating them instead as “ignorable”; and (3) that *if*-clauses function to restrict the context in which they occur for the purpose of evaluating the consequent, and do nothing else. These assumptions interact to derive a trivalent semantics for conditionals from first principles. In this approach, conditionals are not special in any way.

²⁶Undefined antecedents and embedded conditionals may require some additional care.

The temporary-context is a formal implementation of the Ramsey Test, but it does not require any special-purpose pragmatics around “conditional assertion” or the like. Instead, its key features emerge from interaction with features of the semantic and pragmatic theory that are independently motivated by the theory of presupposition, general pragmatic phenomena, and the interpretation of the third value as “ignorable”. To be sure, much empirical, theoretical, logical, and philosophical work remains to be done. Nevertheless, the treatment of *if*-clauses as devices for temporarily modifying the context stands as a promising contender among theories of conditionals in terms of its ability to achieve empirical and theoretical unification and explanatory goals, while also making use of the least possible machinery specific to the theory of conditionals. In addition, it points to a novel generalization that may have broader implications: possibilities outside the local context of interpretation are systematically ignored by the semantic interpretation mechanism.

Compliance with Ethical Standards

There are no conflicts of interest to report. This research did not involve human participants.

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